

Machine Learning in Practice I

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About the course

- Course Title: "Machine Learning in Practice I"
- instructor: antal.jakovac@uni-corvinus.hu
- course objectives
 - ➔ why and when to use Machine Learning (ML)
 - ➔ data handling methods
 - ➔ core ML models before deep learning
 - ➔ how to build an ML project



- about the course
 - ➔ evaluation: written exams (together with exercises)
 - ➔ learning material: course slides, books, podcasts, etc.
 - ➔ books:
 - An Introduction to Statistical Learning
(G. James , D. Witten , T. Hastie , R. Tibshirani, <https://www.statlearning.com/>)
 - Designing machine learning systems (Chip Huyen)
 - ➔ github repository: <https://github.com/ajakovac/ML-in-Practice-I>



- about the course

- ➔ evaluation: written exams (together with exercises)
- ➔ learning material: course slides, books, podcasts, etc.

- ➔ books:

- An Introduction to Statistical Learning
(G. James , D. Witten , T. Hastie , R. Tibshirani, <https://www.statlearning.com/>)

- Design *#git clone project from command line*

- ➔ github repository

```
git clone https://github.com/ajakovac/ML-in-Practice-I.git  
cd ML-in-Practice-I
```

#read the README.md file and follow instructions

Introduction



- why do we need Artificial Intelligence (Machine Learning)?

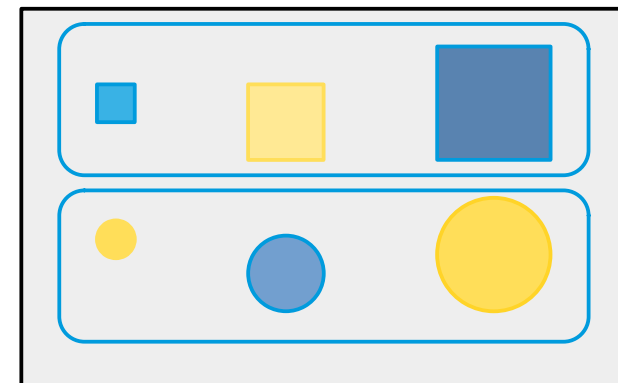
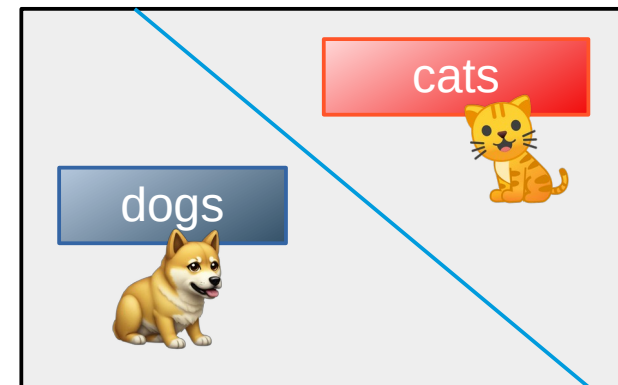
	observation	modelling	computation	conclude/action
historic times	human	human	human	human
machines	machine	human	human	human
computers	machine	human	machine	human
AI	machine	machine	machine	human

- task of AI is not to solve an actual problem, but solve modelling
- we do not know the model it uses → it can be smarter than us!



- Traditional Programming vs. ML
 - ➔ traditional: data + rules → output
 - ➔ ML: data + output → rules
- when to use ML?
 - ➔ Automation of complex tasks → heuristics
 - ➔ Adaptability
 - ➔ Performance on large data
 - ➔ Real-world examples (email spam, face detection, recommendation systems)

- Types of ML; traditionally
 - supervised:
 - show data + labels → learns the assignment
 - examples: classification, regression
 - unsupervised:
 - find patterns in unlabeled data
 - not unique → context, magnitude
 - examples: clustering, dimensionality reduction
 - reinforcement:
 - learn by interacting with the environment
 - not covered in this course





- Types of ML; other points of view
 - task-driven (System I)
 - define a task (classification, prediction, control)
 - show examples of good (eventually bad) solutions
 - train models to optimize for success on that task
 - includes most supervised/reinforcement learning examples
 - data-driven (System II)
 - reveal structures by showing similar data
 - does not need explicit labelling
 - LLM's, autoencoders, similarity learning
 - recently popular (self-supervised learning, foundation models, and retrieval-based systems)



- Typical ML tasks
 - classification
 - face, dog breeds, bird songs classification
 - spam detection, disease diagnosis, sentiment analysis
 - lot of labeled data
 - regression (price prediction, weather forecasting, energy demand)
 - clustering (customer segmentation, document grouping, biological cell types)
 - compression (PCA, visualization, autoencoders, embeddings)



- Typical ML tasks
 - ➔ decision making (robotics, game playing, resource allocation, monitoring)
 - ➔ generation (text, translation, image, data, molecular structure, code)
 - ➔ outlier analysis (email spam, suspicious customer, fraud detection, etc.)
 - ➔ recommendation systems (movies, jobs, code completion, etc.)
 - ➔ ranking/retrieval (search engines, document/data retrieval)



- somewhat overlapping areas; e.g. chemical plant monitoring system
 - ➔ anomaly detection
 - ➔ classification
 - ➔ time-series forecasting
 - ➔ reinforcement learning (control), recommendation systems



- ML algorithms (pre-deep neural networks)
 - Linear regression, logistic probability
 - Bayesian analysis, probability theory
 - Support Vector Machines
 - Decision Trees, ensemble methods
 - k-nearest neighbour method
 - Principal Component Analysis
 - Extreme Learning Machine
- majority of ML applications in production are not DNN!



Tools

- language: Python
- packages:
 - numpy: numerical computing
 - scipy, scikit-learn: classical ML tools
 - pandas: data manipulation
 - matplotlib, seaborn: visualization



Tools

- Python virtual environment:

```
> python -m venv .venv
```

- activate environment:

- Linux/Windows WSL:

```
> source .venv/bin/activate
```

- Windows PowerShell:

```
> .venv\Scripts\Activate
```

- Install necessary packages within the environment

```
> pip install matplotlib numpy scipy scikit-learn pandas seaborn ipykernel
```

- automation:

```
> pip freeze > requirements.txt
```

```
> pip install -r requirements.txt
```



Tools for code development

- notebook services: Jupyter, Google Colab
- advanced editors: Visual Studio Code or similar
- Linux-like environment in Windows: WSL
- docker: containerize your work – like a standalone machine
- communication: API endpoints (Python fastAPI or Flask)

Typical ML workflow



- Typical ML workflow
 - 1) Problem definition
 - 2) Data collection
 - 3) Data cleaning & preprocessing
 - 4) Feature engineering
 - 5) Model selection
 - 6) Training & validation
 - 7) Evaluation
 - 8) Deployment & monitoring



- Typical ML workflow

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- what is the problem we want to solve? (classification, regression, clustering, ...)
- is ML an adequate tool?
- what does success look like? (accuracy? low error? business impact?)
- what are the domain constraints and goals (e.g. medicine data availability, privacy → practical, ethical or technical limits)
- example: predict customer churn or classify handwritten digits



- Typical ML workflow

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- enough data
- sampling methods → diverse data
- labelled data (if supervised)
- example: downloading a CSV of housing prices or collecting user logs



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- 
- missing values, duplicates, outliers
 - normalize, scale, encode categorical features
 - convert raw inputs into usable formats
 - example: fill missing ages with median; scale prices to [0,1]



- Typical ML workflow

- 1) Problem definition
- 2) Data collection
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- 7) Evaluation
- 8) Deployment & monitoring

- 
- create meaningful input features
 - transform or combine raw attributes
 - select or construct new features based on domain knowledge
 - example: from date of birth → compute age; combine “city” and “job type”



- Typical ML workflow

- 1) Problem definition
- 2) Data collection
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- choose a suitable algorithm (e.g. linear regression; SVM; RF; k-NN)
- consider interpretability, complexity, training time
- example: try logistic regression first for classification, then maybe a random forest, or a deep neural network



- Typical ML workflow

- 1) Problem definition
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- 
- split data into training and validation sets (and possibly test)
 - train on one part, validate on another
 - tune hyperparameters (e.g., via cross-validation)
 - example: train a model on 80% of the data, validate on 20%



- Typical ML workflow
 - 1) Problem definition
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Use metrics appropriate for the task:

- classification: accuracy, precision, recall, F1
- regression: MSE, MAE, R^2
- plot confusion matrices, ROC curves, residuals
- example: check precision/recall on spam classification



- Typical ML workflow

- 1) Problem definition
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- integrate the model into production (web app, API, etc.)
- monitor for drift, performance degradation
- retrain if needed (based on new data or feedback)
- example: host model on a server and track prediction accuracy over time

Logics of Intelligence



- Consider a world W consisting of 2 pixels
 $\{\square\square, \square\blacksquare, \blacksquare\square, \dots \blacksquare\blacksquare\}$
- W also has two visible states A and B . We observe that
 - $A \rightarrow \{\square\square, \blacksquare\square, \dots \blacksquare\square\}$
 - $B \rightarrow \{\square\blacksquare, \blacksquare\blacksquare, \dots \blacksquare\blacksquare\}$
- **Step1:** problem definition: tell the state from the pixels
→ classification task



- **Step2-3:** Data collection, cleaning, preprocessing
- in our case: observation → numerics
- the pixels are black-and-white → brightness is a number in $[0,1]$
 - black: 0
 - white: 1
 - grey: somewhere in between
- 2 pixels mean 2 numbers (x_1, x_2)
- *usually Step2-3 is rather tedious...*

- **Step4-5-6:** feature engineering, model selection, training (in practice these are separate steps)
- Let us plot the point pairs corresponding to state A and B

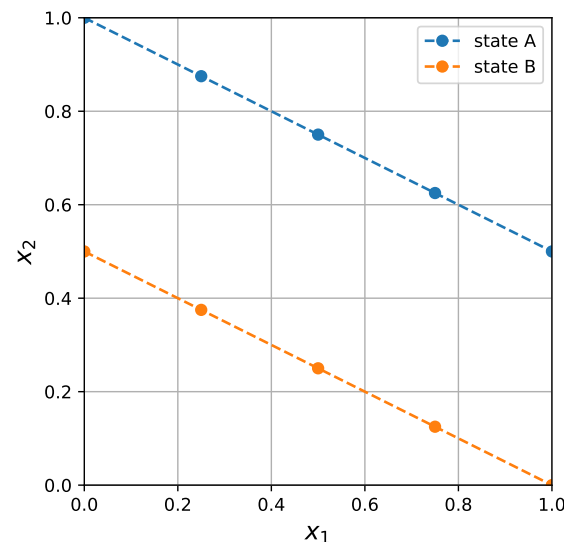
→ the points lie in subspaces (lines)

→ within state A and B

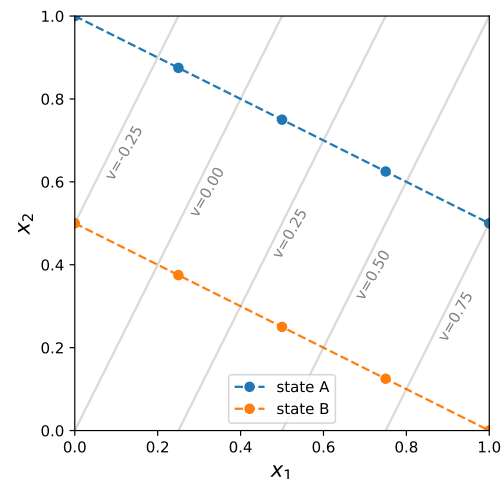
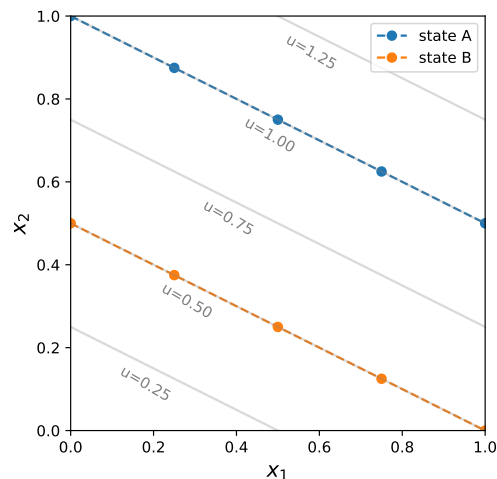
$$\frac{x_1}{2} + x_2 = \begin{cases} 1 & \text{for state A} \\ 0 & \text{for state B} \end{cases}$$

→ worth to introduce new coordinates (features)

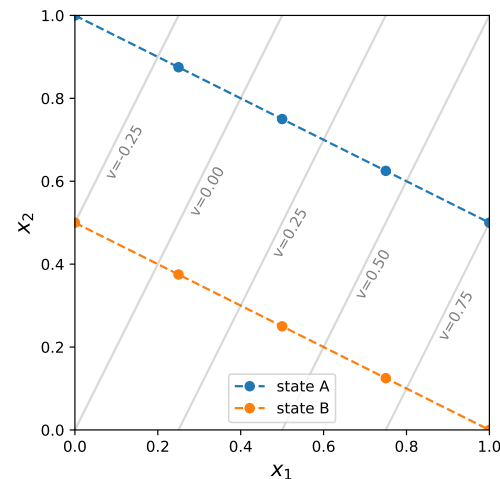
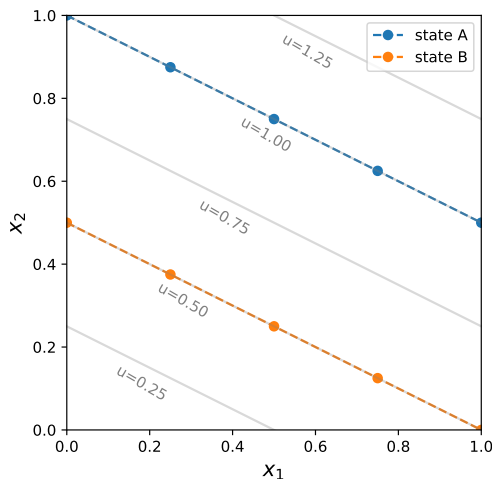
$$u = \frac{x_1}{2} + x_2, \quad v = x_1 - \frac{x_2}{2}$$



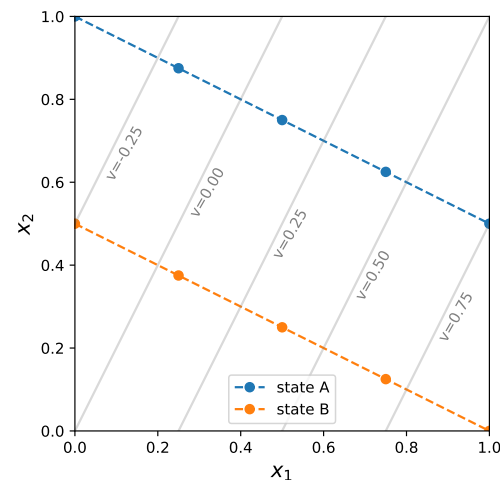
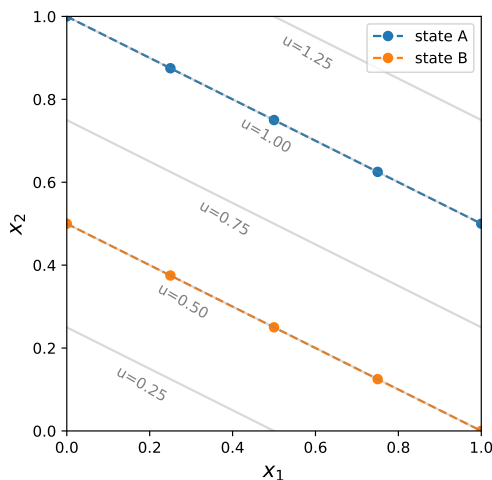
- **Step4-5-6:** feature engineering, model selection, training
- with the new coordinates $x_1 = \frac{2u}{5} + \frac{4v}{5}$, $x_2 = \frac{4u}{5} - \frac{2v}{5}$
- constant u – aligned with states; constant v – changes within a given state



- **Step4-5-6:** feature engineering, model selection, training
- To tell apart state A and B we need only the value of u !
 - ➔ $u = 1.0$ state A
 - ➔ $u = 0.5$ state B
 - ➔ otherwise neither
- solves classification task



- **Step4-5-6:** feature engineering, model selection, training
- To describe a instance belonging to a state we need just v !
 - state A: v in $[-0.5, 0.75]$
 - state B: v in $[-0.25, 1]$
- solves compression,
dimensional reduction





- **Step4-5-6:** feature engineering, model selection, training
- a perfect **feature selection** provides a **coordinate system** best aligned with the equivalence classes of the data
 - there are coordinates that are constant on each data class, and have different values on different classes
(*relevant/selective coordinates* – good for classification)
 - there are coordinates that change within a given data class
(*descriptive/irrelevant coordinates* – good for compression)
- **task of all model building is to find (approximately) these coordinates/features**



- **Step4-5-6:** feature engineering, model selection, training
- in practice we separate the task of coordination
 - ➔ **feature selection:** original features, simple combinations
 - ➔ **model selection:** single out a parametrizable functional space to combine the features (distance, linear- or nonlinear combinations)
 - ➔ **training:** determine the free parameters of the model that fits the data the best



- **Step4-5-6:** feature engineering, model selection, training
- **mind vs data?**
 - data/models importance ratio → data are central importance in real world applications
 - are data enough? do we possess the necessary knowledge?
 - present approach: data engineering is the most important, few fundamental research (cf. LeCun vs Wang in Meta)



- **Step7:** evaluation
- try to estimate how well the original task was solved
 - class reconstruction accuracy
 - robustness, proclivity for failures
 - treatment of outliers
 - business benefits



- **Step8:** deployment and monitoring
- in a real-life application the ML code is used as a product
- requires continuous monitoring
 - ➔ data/environment may change
 - ➔ new classes appear
 - ➔ errors may occur



takehome message: Task of the intelligence is to

find the **relevant features** and **adapt** their values to the task

Organizing ML projects

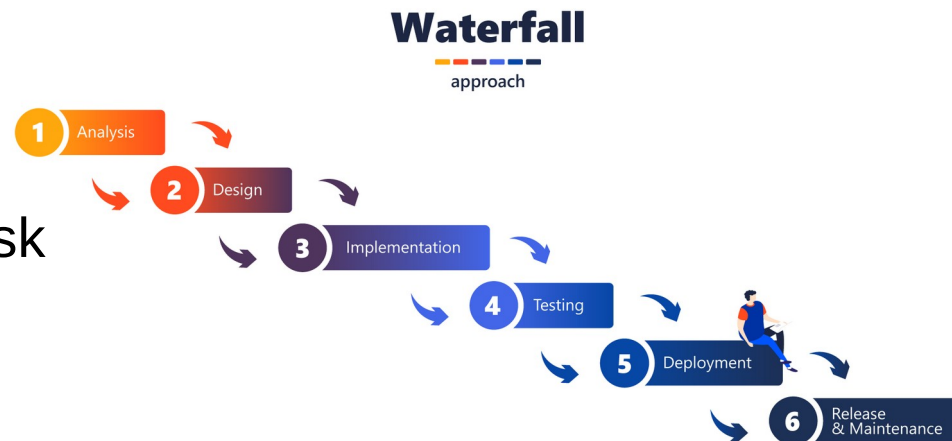
Organizing ML projects

- two popular workflow schemes today: **waterfall** vs. **agile**

- waterfall** workflow:

- clear documentation of the task
- predictable timeline
- inflexible
- late discovery of errors

- good for well-defined projects** (like bridge building)



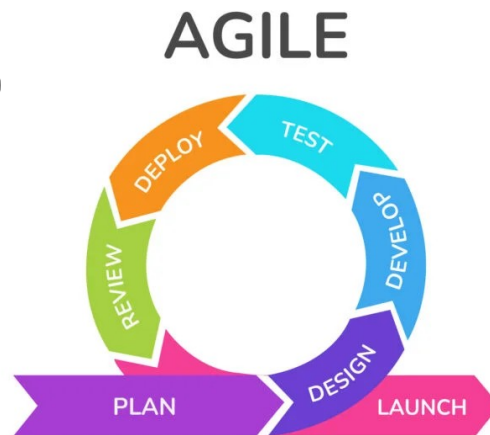
Organizing ML projects

- **agile** workflow: (agile manifesto, description)

- ➔ individuals > processes
 - collaborative vs rigid teams
- ➔ working model > detailed documentation
 - Minimal Viable Product MVP
 - improvement

"If you're not embarrassed by the first version of your product, you've launched too late."

— Reid Hoffman, founder of LinkedIn

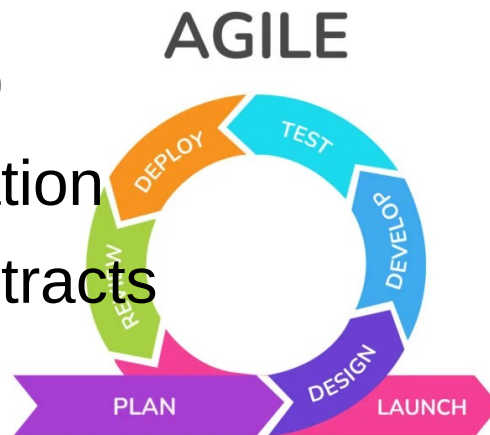


Goal: Solve transportation

- ➔ MVP: Skateboard
- ➔ Next: Bicycle
- ➔ Then: Motorcycle
- ➔ Finally: Car

Organizing ML projects

- **agile** workflow: (**agile manifesto**, **description**)
 - ➔ customer collaboration > contract negotiation
 - customer needs even over signed contracts
 - ➔ responding to change > following a plan
 - original ideas/original task estimates may fail
 - continuous adaptation
 - ➔ launch a “good enough” product



Organizing ML projects

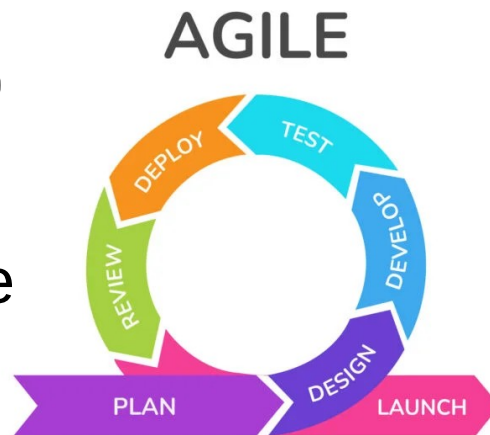
- **agile** workflow: (agile manifesto, description)

- ➔ pros:

- flexible, self-improving, communicative
- customer-centric

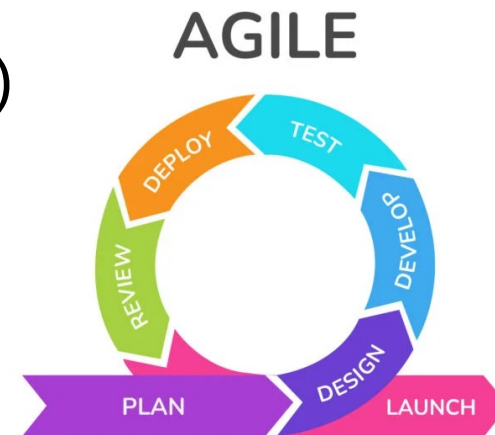
- ➔ cons:

- over-emphasized meetings (“agile theater”)
- over-controlled
- less predictable



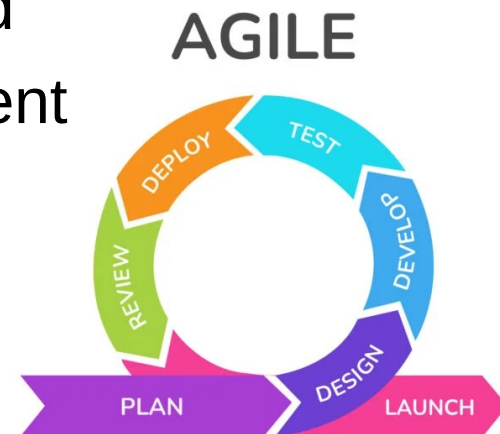
Organizing ML projects

- **agile** workflow: (**agile manifesto**, **description**)
 - ➔ loops, sprints
 - ➔ user interface/user experience (UI/UX)
 - ➔ CI/CD, CI/CD pipelines
continuous integration, continuous deployment
 - ➔ Kanban/Jira: project development visualization
(backlog, to do, in progress, testing, done)



Organizing ML projects

- typical small group agile project
 - ➔ define the goal (user stories), create a backlog (list of tasks)
 - ➔ make repo (github), branches
 - ➔ make MVP, backend API, frontend, testbed
 - ➔ get MVP working, deploy to test environment
 - ➔ collect feedback (team + users)
 - ➔ refine tasks, and directions to proceed





Structure of a professional ML organization

-  Business side
 - ➔ client/stakeholder: business requirements (e.g., “reduce equipment downtime,” “recommend better products”)
 - ➔ Business Analyst / Product Manager (translates business goals into ML problems, acts as the bridge between client and technical teams)
 - ➔ Project/ Product Manager (budgeting, timelines, and resource allocation, Coordinates with business analysts and technical leads to estimate cost)

Production level ML organization

-  Data/model experts
 - ➔ data engineer: build and maintain data pipelines → data collection, cleaning, validating, transformation (ETL/ELT), storing (database, data lake), monitoring
 - ➔ ML engineer/data scientist
 - feature generation
 - design, train, and fine-tune ML models
 - explores and experiments with data, build prototypes
 - performance evaluation

Production level ML organization



Software & Infrastructure

- ➔ software architect: designs overall system, service planning, integration with other services, APIs, databases
- ➔ software developer (coder): implements algorithms, frontend-backend
- ➔ DevOps/MLOps engineer:
 - automates deployment, monitoring, scaling of ML models
 - works with Docker, Kubernetes, CI/CD pipelines, cloud platforms.
 - ensures the ML system is reproducible, reliable, and easy to maintain.

Production level ML organization

-  Operations, Monitoring & Safety
 - ➔ QA / Test Engineer: tests software and ML pipelines (unit tests, integration tests, stress tests), ensures reliability before release
 - ➔ Monitoring & Reliability Engineer: monitors system health after deployment.; tracks data drift, performance degradation, unusual patterns
 - ➔ Security Engineer: ensures compliance with data privacy, protects against adversarial attacks, model leaks.



The Full Pipeline in a Story (user story)

- Client says: “I need early warnings for dangers in a chemical plant.”
- Business Analyst refines: “This means anomaly detection on sensor signals.”
- Data Engineer sets up pipelines to collect sensor data.
- Data Scientist explores, builds a prototype anomaly detector.
- ML Engineer optimizes the model for latency and accuracy.



The Full Pipeline in a Story (user story)

- Software Architect decides how it fits into the production system.
- Developers implement the service around the model.
- DevOps/MLOps deploys it to the cloud with monitoring.
- QA & Monitoring check performance and flag issues.
- Security Engineer ensures compliance and robustness.

Programming code structure

How to organize a (ML) code?

- points of view:
 - ➔ effective code development (also in teams)
 - ➔ scalability
 - ➔ extensibility
 - ➔ easy maintenance



Original approach: monolith program structure

- UI, business logic, data access in one code
- coding logic: *input data* → *transformation* → *resulting data*
- best fits for imperative/procedural languages (like C, Fortran)
 - ➔ program state: mutable variables
 - ➔ program is a chain of commands, changing the state
- natural approach → original programming style

problems:

- hard to trace bugs or change code:
 - ➔ variable change appear in different functions
 - ➔ frequently hidden state change in procedures
 - ➔ in a large code intractable structure
- functions are specific → not reusable
- experience: monolith codes become unmaintainable
(*software complexity crisis in the 1970s–80s*)
- still used for starting a project (MVP)

```
y = 3 #global state variable
```

```
def function(x):  
    x += y  
    y += 1 #changes y tacitly  
    return x
```




different type of solutions:

- object oriented programming
 - data and the corresponding procedures are bundled: "objects" of the world
 - abstraction: hierarchy of classes
 - inheritance: classes are reusable
 - examples: C++, Java

```
class MyClass:  
    def __init__(self, state):  
        self.state = state  
  
    #change of state is a class method  
    def rotate(self, angle):  
        self.state → self.state'(angle)  
        return self
```



- functional programming

- stateless programs, immutable variables
- functions give back new objects (pure functions)
- based on mathematical logic (λ -calculus)
e.g. $\lambda x. x+1$ and their application, e.g. $(\lambda x. x+1) 5 \rightarrow 6$
- basic elemental functions: see next page

```
counter=0
def impure():
    counter+=1 #side effect
    return counter
def pure(x):
    return x+1 #no side effect
```



- basic elemental functions

- map: `map ( $\lambda x. x*2$ ) [1,2,3]  $\rightarrow$  [2,4,6]`

- reduce/fold: `fold (+) 0 [1,2,3]  $\rightarrow$  6`

- filter: `filter even [1,2,3,4]  $\rightarrow$  [2,4]`

- zip: `zip ['a','b','c'] [1,2,3]  $\rightarrow$  [('a',1),('b',2),('c',3)]`

- compose: `(compose f g) x  $\rightarrow$  f(g(x))`

- examples: Haskell, Lisp, F#

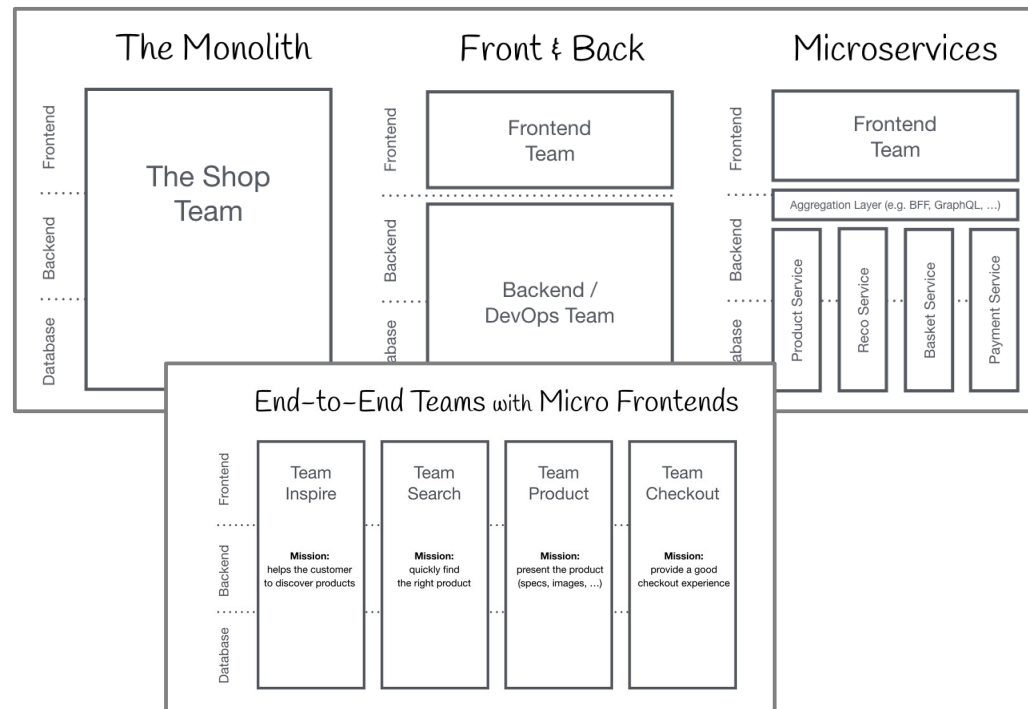
- not popular in the original form, but lot of elements are applied



- mixed (multi-paradigm) languages:
 - ➔ allow all kind of programming style
 - ➔ examples: Python, JavaScript, Rust, Swift,
 - ➔ e.g. Python has map, filter, reduce, based on lists.
 - ➔ Python supports OOP, classes, inheritance, etc.

Other way of deviating from monolithic imperative coding

- monolithic structure
- frontend+backend
- microservices
- micro frontend structure
(c.f. <https://micro-frontends.org/>)



Backend + frontend

- rise of web (1990s) browsers became universal clients
- specialized tools for browser programming (like HTML, CSS, JavaScript)
- task solving (backend) + visualization (frontend) separately
 - ➔ develop separately, different languages, tools
 - ➔ scale BE and FE separately
 - ➔ needs interface management → see next page



- functions designed for intra-code communication, can not be applied any more
- inter-service communication standards:
 - ➔ APIs (application programming interface → synchronous)
 - ➔ queues (message buffer → asynchronous)

- APIs – call server functions (endpoints)
 - ➔ needs continuously running background processes (web server process), listening to input channels (ports)
 - ➔ endpoints: well defined input and output formats, usually in json
 - ➔ communication → REST API: get, put/patch, post, delete
 - ➔ in Python: Flask, fastAPI
 - ➔ can be asynchronous, too

#fastAPI snippet

```
from fastapi import FastAPI
app = FastAPI()
@app.get("/")
def read_root():
    return {"message": "Hello ✨"}
```

#start service

```
uvicorn main:app
```

#call service

```
open http://127.0.0.1:8000, or
curl http://127.0.0.1:8000/
results {"message": "Hello ✨"}
```




- queues: async send message
 - ➔ message broker runs in background, it stores messages
 - ➔ worker processes deliver messages for subscribers/consumers
 - ➔ data plain text or json
 - ➔ metadata + payload
 - ➔ in Python: queue + threading

#queue application example

```
import queue  
q = queue.Queue()
```

#put data to queue

```
q.put(item)
```

#read data from queue

```
item = q.get()
```

#... process data ...

```
q.task_done()
```



Microservices + frontend

- good scaling properties (e.g. web users) → requiring independence of elementary tasks became popular
- lot independent, narrowly scoped (micro) services
 - separate development (languages, tools, teams), deployment
 - separate scaling
 - resilience: bugs are confined, not affect the whole system
- examples Netflix, Amazon → hype in 2010s



- new challenge: how to organize a lot of services
- microservice management: docker, Kubernetes

#example Dockerfile

```
FROM python:3.12-slim
```

```
WORKDIR /app
```

```
COPY main.py .
```

```
RUN pip install fastapi uvicorn
```

```
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

#example docker-compose.yml

```
services:
```

```
  service1:
```

```
    build: ./service1
```

```
    container_name: service1
```

```
    ports:
```

```
      - "8001:8000"
```

```
  service2:
```

```
    build: ./service2
```

```
    container_name: service2
```

```
    ports:
```

```
      - "8002:8000"
```

```
docker-compose up --build
```



- new challenge: how to organize a lot of services
- CI/CD pipelines:
 - ➔ continuous improvement
 - ➔ continuous deployment
 - ➔ team communication
- example: github



- new challenge: how to organize a lot of services
- CI/CD pipelines:
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 - ➔ team communication
- example: github

```
#example .github/workflows/ci.yml
name: CI
on:
  push: branches: ["main"]
  pull_request: branches: ["main"]
jobs:
  build-and-test:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt
          pip install pytest
```



- new challenge: how to organize a lot of services

- CI/CD pipelines:

- continuous integration
- continuous deployment
- team communication

- example: github

#practical example: git clone project

```
git clone https://github.com/ajakovac/ML-in-Practice-I.git  
cd ML-in-Practice-I
```



- despite all advantages microservices have drawbacks
 - ➔ operational complexity → manage hundreds of services
 - ➔ network calls → latency, performance issues
 - ➔ interprocess debugging is hard



- Current trends (2020s → now)
 - ➔ modular monolith applications
 - ➔ microservices where they make sense
 - ➔ functions-as-a-service (FaaS)
 - in cloud services like AWS, Azure, Google cloud
 - locally they seem like functions
 - behave like services when deployed
 - run on demand



- Current trends
 - ➔ modular n
 - ➔ microserv
 - ➔ functions-
 - in clou
 - locally
 - behave
 - run on

#example in AWS

#lambda_function.py

```
def lambda_handler(event, context):  
    name = event.get("name", "my_name")  
    return { "statusCode": 200,  
            "body": f"Hello, {name}! ✨"
```

zip function.zip lambda_function.py

```
aws lambda create-function \  
  --function-name helloLambda \  
  --runtime python3.12 \  
  --role arn:aws:iam::<YOUR_ACCOUNT_ID>:role/lambda-ex-role \  
  --handler lambda_function.lambda_handler \  
  --zip-file fileb://function.zip
```

curl -X POST \
 -H "Content-Type: application/json" \
 -d '{"name": "MLcourse"}' \
 https://abcd1234.execute-api.us-east-1.amazonaws.com/default/hello

Code infrastructure → software architecture

- use few necessary microservices → clear task definition
- use lambda for simple services
- containerize microservices → portability, environmental stability
- use container management → docker (Kubernetes)
- use CI/CD management → github or similar

Data management

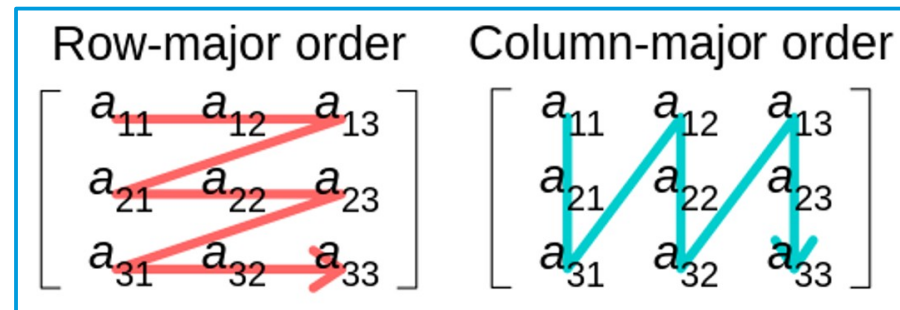


data sources:

- user input (text, image, video, etc.) – often not well formed, errors
- system generated (logs, measurements, predictions) – well formed data, easy-to-use
- own data, own clients (first-party), other companies, their own clients (second-party), other companies not own clients (third-party – e.g. internet usage, traffic, etc)

data formats

- CSV (comma separated values)
 - text, human readable
 - tabular, row-major
 - fast to get data belonging to the same example
- Parquet
 - binary – more compact, but not human readable
 - tabular, column major
 - fast to get data belonging to the same feature



data formats

- CSV (comma-separated values)

- text, human-readable
- tabular, row-oriented
- fast to get data out

- Parquet

- binary – more compact
- tabular, column-oriented
- fast to get data out

#example books.csv

title,author,publisher,year

The Hobbit,J.R.R. Tolkien,George Allen & Unwin,1937

The Fellowship of the Ring,J.R.R. Tolkien,George Allen & Unwin,1954

Nineteen Eighty-Four,George Orwell,Secker & Warburg,1949

Brave New World,Aldous Huxley,Chatto & Windus,1932

The Catcher in the Rye,J.D. Salinger,"Little, Brown and Company",1951

#read data using pandas

import pandas as pd

df = pd.read_csv("books.csv")

print(df)

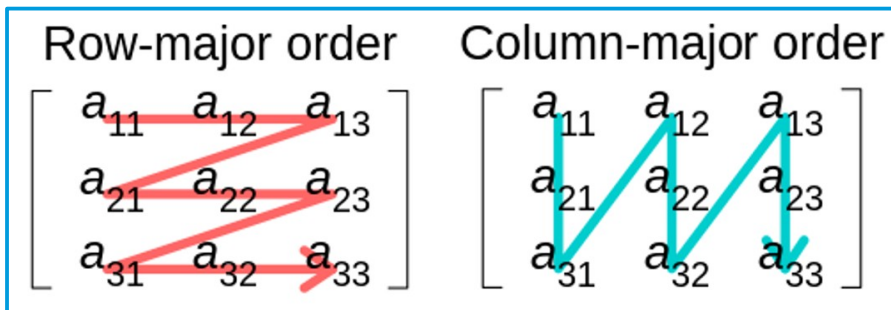
#output

	title	author	publisher	year
0	The Hobbit	J.R.R. Tolkien	George Allen & Unwin	1937
1	The Fellowship of the Ring	J.R.R. Tolkien	George Allen & Unwin	1954
2	Nineteen Eighty-Four	George Orwell	Secker & Warburg	1949
3	Brave New World	Aldous Huxley	Chatto & Windus	1932
4	The Catcher in the Rye	J.D. Salinger	Little, Brown and Company	1951

data formats

- CSV (comma separated values)

- text, human readable
- tabular, row-major
- fast to get data belonging to the same example



- Parquet

- binary – more compact
- tabular, column-major
- fast to get data belonging to the same column

```
#store in parquet – needs fastparquet to be installed
df.to_parquet("books.parquet", engine="fastparquet", index=False)
-----
#store in parquet – needs fastparquet to be installed
df_parquet = pd.read_parquet("books.parquet", engine="fastparquet")
```



data formats:

- JSON (JavaScript Object Notation) → today's standard
 - hierarchical structure
 - text, human readable
 - key-value pairs
 - in Python: dict
 - supported by all languages

data formats:

- JSON (JavaScript Object Notation)

- hierarchical
- text, human-readable
- key-value pairs
- in Python: `json` module
- supported by most programming languages

```
#example books.json
```

```
[  
  {  
    "title": "The Hobbit",  
    "author": "J.R.R. Tolkien",  
    "publisher": "George Allen & Unwin",  
    "year": 1937  
  },  
  {  
    "title": "The Fellowship of the Ring",  
    "author": "J.R.R. Tolkien",  
    "publisher": "George Allen & Unwin",  
    "year": 1954  
  },  
  ...  
]
```

```
#read with pandas
```

```
df = pd.read_json("books.json")
```



data models:

- for more complicated data, and if queries are required, storing data in simple files is not enough
- instead: databases with dedicated database handlers
 - ➔ relational (e.g. Sqlite)
 - ➔ nonrelational (e.g. Redis, Mongo)

data models:

- relational
 - tabular logic: main tables, subtables for standardization, index tables for faster query
 - strict schemas
 - query language for data retrieval → SQL (structured query language): declarative, specify the result, not the algorithm
 - **pros**: simple logic, widely used
 - **cons**: abundant schema systems, lots of superficial data, tedious to introduce new features, complicated table structure

data models:

- relational

- tabular logic
faster query
- strict schema
- query language
declarative,
- **pros:** simple
- **cons:** abundant
introduce n

#example sqlite application – built in Python package

```
import sqlite3
```

#create/connect database

```
conn = sqlite3.connect("Data/books.db")
```

```
cur = conn.cursor()
```

#create tables

```
cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS publishers (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    name TEXT NOT NULL
```

```
)
```

```
""")
```

```
cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS books (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    title TEXT NOT NULL,  
    author TEXT NOT NULL,  
    year INTEGER,  
    publisher_id INTEGER,  
    FOREIGN KEY (publisher_id) REFERENCES publishers(id)
```

```
)
```

```
""")
```

data models:

- relational

- tabular logic
faster query
- strict schema
- query language
declarative
- **pros:** simple
- **cons:** abundant
introduce n

```
#insert a publisher
```

```
cur.execute("INSERT INTO publishers (name) VALUES (?)", ("George Allen & Unwin",))
```

```
publisher_id = cur.lastrowid
```

```
# Insert a book linked to that publisher
```

```
cur.execute("INSERT INTO books (title, author, year, publisher_id) VALUES  
(?, ?, ?, ?)", ("The Hobbit", "J.R.R. Tolkien", 1937, publisher_id))
```

```
#commit changes
```

```
conn.commit()
```

```
-----
```

```
# Query join
```

```
cur.execute("""
```

```
SELECT b.title, b.author, p.name AS publisher
```

```
FROM books b
```

```
JOIN publishers p ON b.publisher_id = p.id
```

```
""")
```

```
print(cur.fetchall())
```

```
conn.close()
```

```
-----
```

```
#output
```

```
[('The Hobbit', 'J.R.R. Tolkien', 'George Allen & Unwin')]
```

data models:

- NoSQL: no SQL → not only SQL
 - pros: no schemas → flexibility, scalability, treat of unstructured data
 - cons: storage capacity, performance trade-offs, data chaos
 - document stores: data are stored as a JSON (string)
 - query can be a more difficult task
 - examples: mongoDB, CouchDB – create own index tables for effectivity
 - key-value pairs: for fast query (example: RedisDB, DynamoDB)
 - graph databases: concentrate on relations (nodes and edges)
(Examples: Neo4j, Amazon Neptune, ArangoDB)



data models:

- NoSQL: no S

- ➔ pros: no sc

- ➔ cons: stora

- ➔ document s

- query ca

- examples

- ➔ key-value p

- ➔ graph data

(Examples: N

```
#install mongodb-image and pymongo
```

```
docker run -d --name mongodb -p 27017:27017 -v ~/my_data/mongo:/data/db mongo:7.0
```

```
pip install pymongo
```

```
-----
```

```
#runs a service in the background – start/stop it
```

```
docker start/stop mongodb
```

```
-----
```

```
#install mongodb and pymongo
```

```
from pymongo import MongoClient
```

```
# 1. Connect to local MongoDB
```

```
client = MongoClient("mongodb://localhost:27017/")
```

```
# 2. Create (or use existing) database
```

```
db = client["library"]
```

```
# 3. Create collections
```

```
publishers = db["publishers"]
```

```
books = db["books"]
```

```
# 4. Clear old data (for demo purposes)
```

```
publishers.delete_many({})
```

```
books.delete_many({})
```

data

- No

5. Insert publishers

```
pub_allen = publishers.insert_one({"name": "George Allen & Unwin"}).inserted_id
pub_secker = publishers.insert_one({"name": "Secker & Warburg"}).inserted_id
pub_chatto = publishers.insert_one({"name": "Chatto & Windus"}).inserted_id
pub_little = publishers.insert_one({"name": "Little, Brown and Company"}).inserted_id
```

6. Insert books with publisher references

```
books.insert_many([
    {"title": "The Hobbit", "author": "J.R.R. Tolkien", "year": 1937, "publisher_id": pub_allen},
    {"title": "The Fellowship of the Ring", "author": "J.R.R. Tolkien", "year": 1954, "publisher_id": pub_allen},
    {"title": "Nineteen Eighty-Four", "author": "George Orwell", "year": 1949, "publisher_id": pub_secker},
    {"title": "Brave New World", "author": "Aldous Huxley", "year": 1932, "publisher_id": pub_chatto},
    {"title": "The Catcher in the Rye", "author": "J.D. Salinger", "year": 1951, "publisher_id": pub_little},
])
```

7. Query: find all books and join publisher info manually

```
for book in books.find():
    publisher = publishers.find_one({"_id": book["publisher_id"]})
    print(f"{book['title']} by {book['author']} "
          f"was published by {publisher['name']} in {book['year']}")
```




data storage:

- data warehouse:
 - stores historical data → structured, cleaned, integrated
 - optimized for analysis, reporting → few but very large, complex queries
 - ETL (extract, transform, load)
 - OLAP (online analytic processing)
 - examples: historical sales data, stock price data
 - solutions: Amazon Redshift, Google BigQuery



data storage:

- database:
 - stores current, operational data.
 - used for day-to-day transactions (insert, update, delete).
 - optimized for fast read/write, many small simple queries
 - example: an e-commerce database storing users, orders, payments
 - OLTP (Online Transaction Processing)
 - solutions: MySQL, PostgreSQL, Oracle DB, MongoDB, DynamoDB



data storage:

- data lake:
 - stores data in raw form → future-proof
 - application goal not needed to be specify
 - cheap and scalable storage (HDFS, Amazon S3, Azure Data Lake Storage, Google Cloud Storage)
 - schema-on-read solutions (ELT, extract, load, transform) → processing engines (Kafka, TensorFlow, Athena)
 - example: log storage, IoT data storage



data sampling: collecting all data usually not feasible

- nonprobability sampling → focus on data collection; prone to selection bias
 - convenience sampling: take all what is available
 - snowball sampling: start with a small set, and follow its descendants
 - judgement sampling: experts say what to collect
 - quota sampling: collect the same number of data from different areas (like age groups)
 - example of bias: psychology experiments (university students), sentiment analysis (wordy people), medical data (unsuccessful treatments)

data sampling:

- probability sampling → needs a data model → biased?
 - stratified sampling: collect the same amount of data per class
 - weighted sampling: take into account data scaled with probability
 - in choice: class A 25%, class B 75% → consider A samples 3 times
 - in training: give weight for underrepresented classes (see later)
 - reservoir sampling: keep the first k sample, and replace it randomly
 - importance sampling

$$E_P[x] = \sum_x x P(x) = \sum_x x P \frac{P(x)}{Q(x)} Q(x) = E_Q \left[x P \frac{P(x)}{Q(x)} \right]$$



data augmentation:

- if there are not enough data, or not representative enough, we can transform data to generate new ones (symmetries)
 - ➔ rotation, scaling
 - ➔ perturbation, noising
 - ➔ texture or style change
- data generation: simulations in artificial environment (e.g. self-driving cars)



labelling:

- for classification we should collect examples in different classes
- problems:
 - ➔ how unambiguous are the labels? → multi-label classes
 - ➔ how precise are the labels? → no database is 100% correct!
 - ➔ what to do with missing labels? → label functions, semi-supervised, etc.
- hand labelled data, naturally labelled data (context is given), automatically generated data

Monitoring and evaluation

How can we assess the success of the ML results?

- not unique! depends on the problem on hand
 - ➔ example: percentage of correctly classified images
 - ➔ if the probability of belonging to class A is 99%, then the simple method saying that *all* elements belong to A is 99% accurate!
 - ➔ clearly not good, if the task is to find the 1% not-A class!



- baselines: how accurate are simple methods and other approaches?
 - random baseline: choose class randomly → $1/N$ probability for balanced
 - zero rule: all belonging to one class → high probability for imbalanced
 - simple heuristic → use a simple proxy (feature)
 - human experts → how well do humans perform?
 - other ML methods

- confusion matrix

- correct classes vs. estimated classes
- works for multi-class evaluation
- in probabilistic sense joint probability: $C_{ij} = P(\text{predicted}_i, \text{actual}_j)$

	Actual Yes	Actual No	Predicted Total
Predicted Yes	TP=0.510	FP=0.038	PP=0.548
Predicted No	FN=0.032	TN=0.420	PN=0.452
Actual Total	AP=0.542	AN=0.478	1.0

- correctness measures for 2-class evaluation

- *accuracy*: prob. of choosing the correct class: $(TP+TN)$

- *precision*: conditional for Predicted Yes: $P(\text{actual}_p | \text{predicted}_p) = \frac{TP}{PP}$

- *recall*: conditional for Actual Yes: $P(\text{predicted}_p | \text{actual}_p) = \frac{TP}{AP}$

- *type I error*: probability of false alarm: $P(\text{predicted}_p | \text{actual}_N) = \frac{FP}{AN}$

- *type II error*: probability of missing signal: $P(\text{predicted}_N | \text{actual}_p) = \frac{FN}{AP}$



- controlled case → measure does not matter
 - probability of class A or B is 50%, predict symmetrically
 - accuracy=98%
 - for class A: precision=98%, recall=98%, type II error=2%
 - for class B: precision=98%, recall=98%, type I error=2%

	Actual Yes	Actual No	Predicted Total
Predicted Yes	TP=0.49	FP=0.01	PP=0.5
Predicted No	FN=0.01	TN=0.49	PN=0.5
Actual Total	AP=0.5	AN=0.5	1.0

- extreme cases
 - classify everything as class A, but probability of class A is 99%
 - accuracy=99%
 - for class A: precision=99%, recall=100%, type II error=0%
 - for class B: precision=?, recall=0%, type I error=100%

	Actual Yes	Actual No	Predicted Total
Predicted Yes	TP=0.99	FP=0.01	PP=1.0
Predicted No	FN=0.0	TN=0.0	PN=0.0
Actual Total	AP=0.99	AN=0.01	1.0

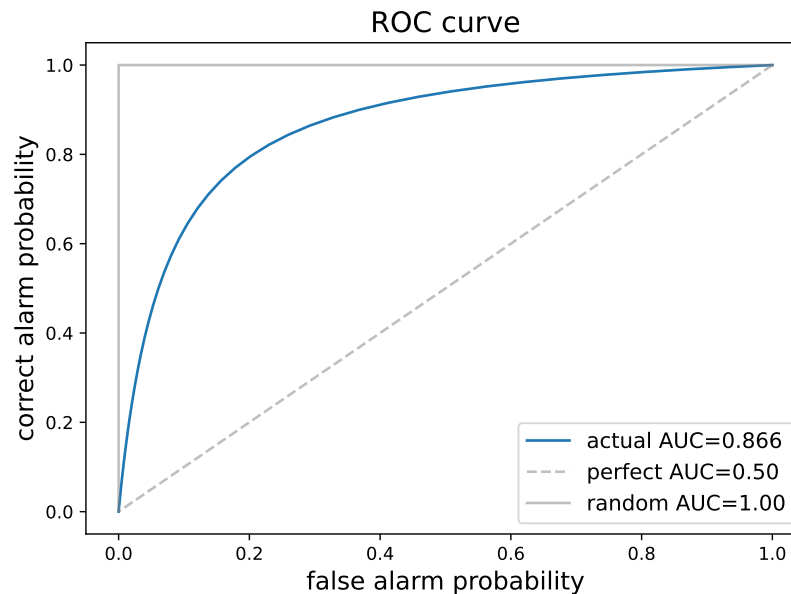
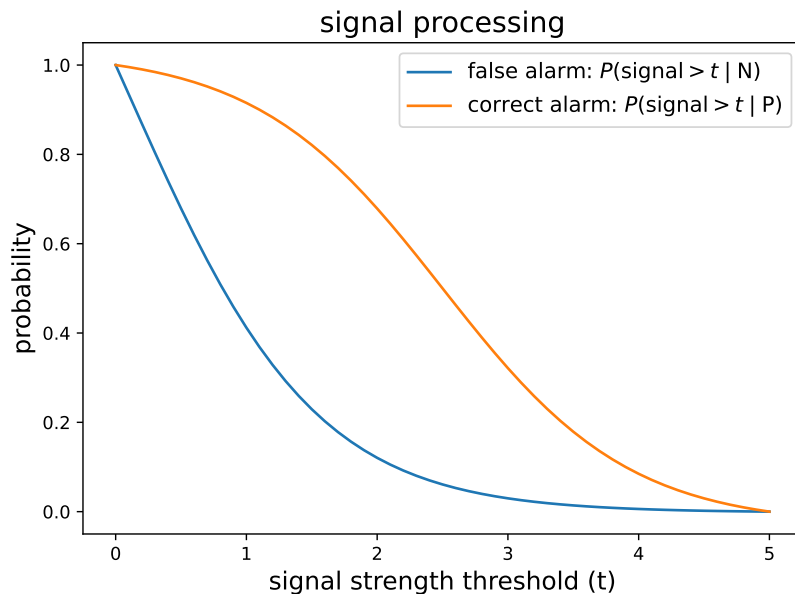
- extreme cases
 - classify everything randomly, but probability of class A is 98%
 - accuracy=50%
 - for class A: precision=98%, recall=50%, type II error=50%
 - for class B: precision=2%, recall=50%, type I error=50%

	Actual Yes	Actual No	Predicted Total
Predicted Yes	TP=0.49	FP=0.01	PP=0.5
Predicted No	FN=0.49	TN=0.01	PN=0.5
Actual Total	AP=0.98	AN=0.02	1.0



- effect of a threshold on the prediction
 - in lot of cases Yes/No decision comes from comparing a signal with a threshold → e.g. smoke detector signal vs. danger level
 - ROC (receiver operating characteristics)
 - $P(\text{predicted}_P | \text{actual}_P)$ vs. $P(\text{predicted}_P | \text{actual}_N)$
 - recall vs. type I error
 - correct alarm probability vs. false alarm probability
 - AUC: area under curve

- effect of a threshold on the prediction



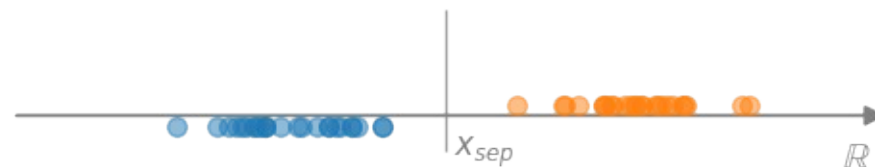
Data modelling

- task: find out the selective/descriptive features
- depends both on the dataset and the context
- selective: constant on a class, descriptive: changes within a class
class depends on the context
- finding the selective coordinates is the main challenge
- we start with simple 1D classification problems, and generalize to several dimensions

One dimensional dataset classification

One dimensional dataset

- Simplest basic case: characterization of one-dimensional data
 - data $\rightarrow x \in \mathbb{R}$ (numbers for analytic characterization)
- Separation of two 1D sets
 - assume that in given sections belong (approximately) to a single class
 - basic case: one section $\rightarrow A$, other section $\rightarrow B$
 - if the two sets are separated (hard margin) $\left\{ \begin{array}{l} x \in A \Leftrightarrow x < x_{sep} \\ x \in B \Leftrightarrow x > x_{sep} \end{array} \right\}$
 - overlapping case? $\rightarrow$ no perfect solution



One dimensional dataset

- overlapping regions → no perfect solution (heuristics)

→ choose x_{sep} somewhere in the overlapping region (eg. midpoint)

→ define an error function in the overlapping region, e.g.

→ $p=1$ (“hinge loss”) → median, $p=2$ → mean

$$L_p(x_{sep}) = \sum_{x \in \text{overlap}} |x - x_{sep}|^p$$

→ we can weight the points



$$0 = \frac{d}{dx_{sep}} \sum_{i=1}^N (x_i - x_{sep})^2 = 2 \sum_{i=1}^N (x_{sep} - x_i) \Rightarrow x_{sep} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\sum_{i=1}^N |x_i - x_{sep}| = \sum_{x_i > x_{sep}} (x_i - x_{sep}) + \sum_{x_i < x_{sep}} (x_{sep} - x_i) = (N_{<} - N_{>}) x_{sep} + \sum_{x_i > x_{sep}} x_i - \sum_{x_i < x_{sep}} x_i \Rightarrow N_{<} = N_{>}$$

- Another approach: treat the problem as a regression

→ define a function $f: \mathbb{R} \rightarrow \mathbb{R}$, that
$$\begin{cases} f(x) < 0 \Leftrightarrow x \in A \\ f(x) > 0 \Leftrightarrow x \in B \end{cases}$$

→ a classification can always be translated to regression

→ simplest: linear relation $f(x) = ax + b$

→ probabilistic interpretation: $p(x) = \frac{e^{f(x)}}{1 + e^{f(x)}} \in [0, 1], \quad p(0) = 0.5$

logistic map, probability of belonging to class B

→ loss function:
$$L(a, b) = \sum_x (p(x) - I_{x \in B})^2, \quad I_{x \in B} = 1 \text{ if } x \in B, \text{ else } 0$$

One dimensional dataset

- Another approach: tree

- define a function $f: \mathbb{R} \rightarrow \mathbb{R}$

- a classification can always be

- simplest: linear relation

- probabilistic interpretation

logistic map, probability of belonging to class B

- loss function:
$$L(a, b) = \sum_x (p(x) - I_{x \in B})^2, \quad I_{x \in B} = 1 \text{ if } x \in B, \text{ else } 0$$

```
# Python function minimization
import numpy, matplotlib
def P(p, x):
    a, x0 = p
    return np.exp(a*(x-x0))/(1+np.exp(a*(x-x0)))

def loss(p, A, B):
    loss = np.sum([P(p, a)**2 for a in A])
    loss += np.sum([(P(p, b)-1)**2 for b in B])
    return loss

p0 = [1, 1] # initial guess
params = (A, B) # values of a, b
res = minimize(loss, p0, args=params)

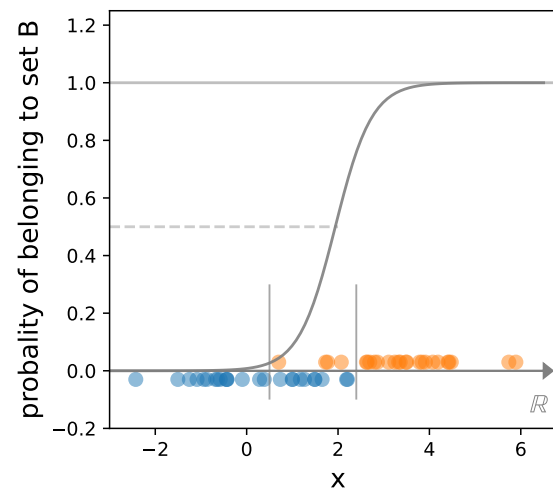
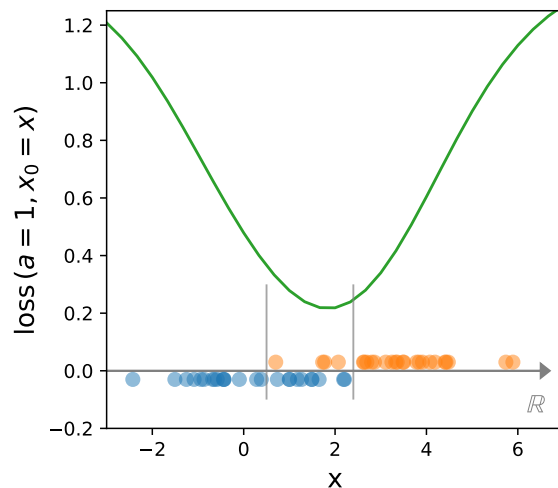
a, x0 = res.x
```


One dimensional dataset

- Another approach: treat the problem as a regression

- ➔ logistic map, probability of belonging to class B

$$p(x) = \frac{e^{ax+b}}{1+e^{ax+b}}$$



Probability and information theory basics

Practical **probability theory** (adapted to ML):

- possible outputs of an event: sample space $\omega \in \Omega$
- **features** ($\rightarrow$ random variables): $X: \Omega \rightarrow \mathbb{R}^n$ functions, $x = X(\omega)$
- observe a list of events: $(\omega_1, \omega_2, \dots, \omega_N)$
- **probability**: $P(X \in U \subset \mathbb{R}^n) = \frac{1}{N} \sum_{i=1}^N \delta_{X(\omega_i) \in U} \rightarrow$ **histogram**
- **distribution**: $p_X(x) = P(X \in \{x\})$
- *probability density*: $p_X(x) = \frac{P(X \in [x, x+dx])}{|dx|}$

Indicator function

$$\delta_b = \begin{cases} 1 & \text{if } b = \text{True} \\ 0 & \text{if } b = \text{False} \end{cases}$$

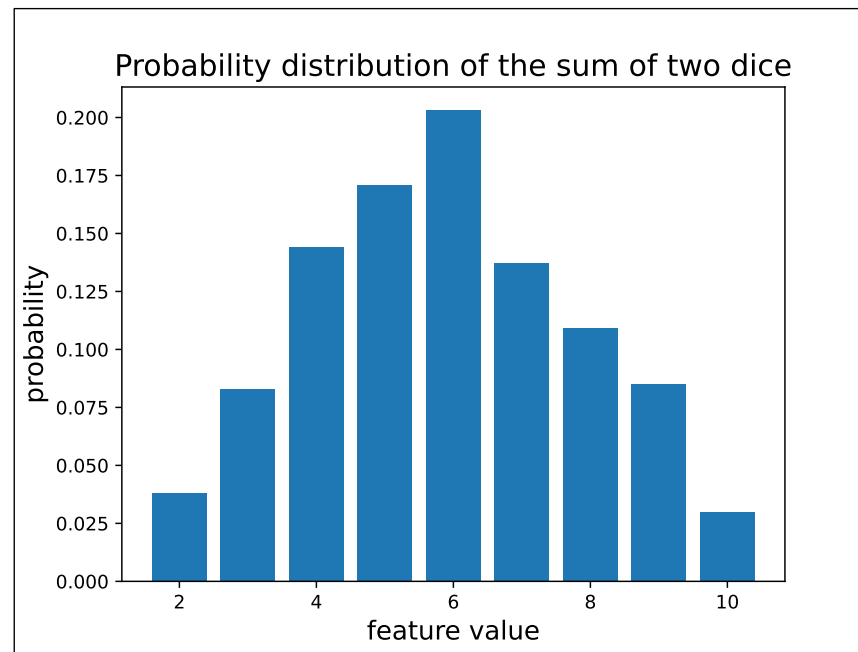


example of 2 dice thrown N times

```
# generate data
Ndata = 1000
data = np.random.randint(1,6,size=(Ndata,2))
feature = data[:,0] + data[:,1]

# measure histogram
fmax = int(max(feature))
fmin = int(min(feature))
bins = np.zeros(fmax-fmin+1)
for f in feature:
    bins[int(f-fmin)] += 1
bins/=Ndata

# show result
fig,ax = plt.subplots()
ax.bar(np.arange(fmin,fmax+1),bins)
ax.set_xlabel('feature value', fontsize=14)
ax.set_ylabel('probability', fontsize=14)
```



Joint probability distributions

- the features can have several components $\rightarrow XY = (X, Y): \Omega \rightarrow \mathbb{R} \times \mathbb{R}$
- their distribution is called joint probability distribution: $p_{XY}(x, y)$
- if we are interested only on one component: $p_X(x) = \sum_{y \in Y} p_{XY}(x, y)$
- for independent features the value of x does not depend on the value of y
$$p_{XY}(x, y) = p_X(x) p_Y(y)$$

$$p_X(x) = \frac{1}{N} \sum_{i=1}^N \delta_{X(\omega_i)=x} = \frac{1}{N} \sum_{i=1}^N \sum_{y \in Y(\Omega)} \delta_{X(\omega_i)=x; Y(\omega_i)=y} = \sum_{y \in Y(\Omega)} p_{XY}(x, y)$$



Information theory: statistical relation of measurements

- assume we have measured a lot $N \rightarrow \infty$
- allows simplifications $\rightarrow$ Stirling formula: $\ln(n!) \approx n \ln n + \dots$
- measure of the uncertainty / unpredictability / impurity of a probability distribution $\rightarrow$ **Shannon entropy**



Shannon entropy: measure of variability of a distribution

- perform N measurements with n possible outputs (bins)
- count the number of results in each bin: $\{N_1, N_2, \dots, N_n\}$
- how many realizations is possible for a given $\{N_1, N_2, \dots, N_n\}$?
- Shannon entropy: log of this number over N
- *equivalent to the entropy concept in physics: number of microstate representation of a macrostate*

Shannon entropy:

- take n bins, N independent measurements
- we obtain $\{N_1, N_2, \dots, N_n\} \rightarrow$ probability $p_i = \frac{N_i}{N}$
- how many ways can this distribution be realized?
(*measurements within a bin are indistinguishable*)
- binomial – multinomial expression

$$\{N_1, N_2\} \rightarrow \binom{N}{N_1} = \frac{N!}{N_1! N_2!} \qquad \{N_1, N_2, \dots, N_n\} \rightarrow \frac{N!}{N_1! N_2! \dots N_n!}$$

Shannon entropy:

- information theory: $N_i \rightarrow \infty$ for all parts
- Stirling formula, using $N = \sum_{i=1}^n N_i$

$$\log \frac{N!}{N_1! N_2! \dots N_n!} \approx N \log N - \sum_{i=1}^n N_i \log N_i = -N \sum_{i=1}^n p_i \log p_i$$

- Shannon entropy:

$$H = \lim_{N_i \rightarrow \infty} \frac{1}{N} \log_2 \frac{N!}{N_1! N_2! \dots N_n!} = - \sum_{i=1}^n p_i \log_2 p_i$$



Shannon entropy: $H = - \sum_{i=1}^n p_i \log_2 p_i$

- always positive or zero $H \geq 0$
- minimal for a pure sample $p_1=1, p_{i>1}=0 \Rightarrow H=0$
- maximal for uniform sample $p_i=\frac{1}{N} \Rightarrow H=\log_2 N$
- convex

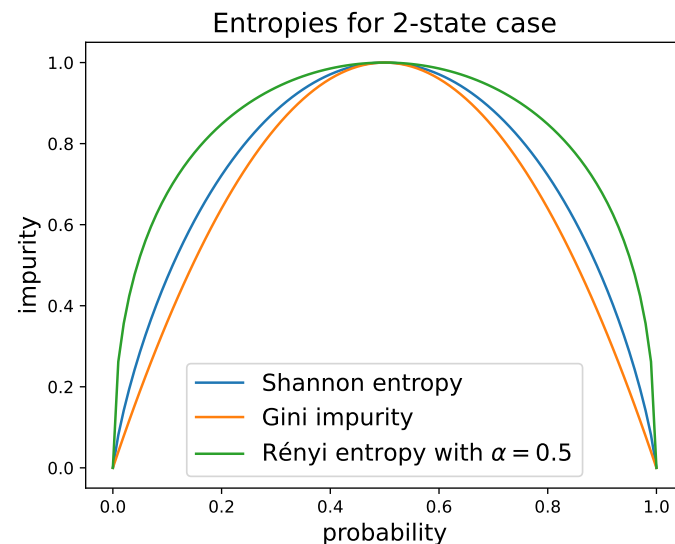
$$\text{if } E = \sum_i E_i P_i \text{ fixed} \Rightarrow p_i \sim e^{-\beta E_i}$$

Other entropy formulae:

- we assumed independence of measurement bins
what if they are not (non-additive systems)
- general definition using properties
(positive, convex)
- examples:

→ Rényi-Tsallis entropy $H_\alpha = \frac{1}{1-\alpha} \log \left(\sum_{i=1}^n p_i^\alpha \right)$

→ Gini impurity $G = 1 - \sum_{i=1}^n p_i^2$



Shannon entropy and amount of information

- we have n events with probabilities $\{p_1, p_2, \dots, p_n\}$
- define **information** as the **minimal number of questions** needed to find out which one happened? → *twenty questions game*
- **answer**: Shannon entropy!
 - one question → one “bit” of information
 - Shannon entropy = information content
 - for uniform probabilities logarithmic search, in general Huffman coding

Shannon entropy and amount of information – technical slide

- coding: a binary code corresponds an interval in $[0,1)$

$$(b_1, b_2, \dots, b_k) \Rightarrow \left[\sum_{i=1}^k b_i 2^{-b_i}, \sum_{i=1}^k b_i 2^{-b_i} + 2^{-k} \right]$$

$$101 \Rightarrow (1, 0, 1) \Rightarrow \left[\frac{1}{2} + \frac{1}{8}, \frac{1}{2} + \frac{1}{8} + \frac{1}{8} \right] = \left[\frac{5}{8}, \frac{6}{8} \right]$$

- can be placed non-overlapping if $\sum_{i=1}^k 2^{-l_i} \leq 1$ (Kraft–McMillan theorem)

- choose code lengths $l_i = \lceil -\log_2 p_i \rceil \rightarrow$ non-overlapping

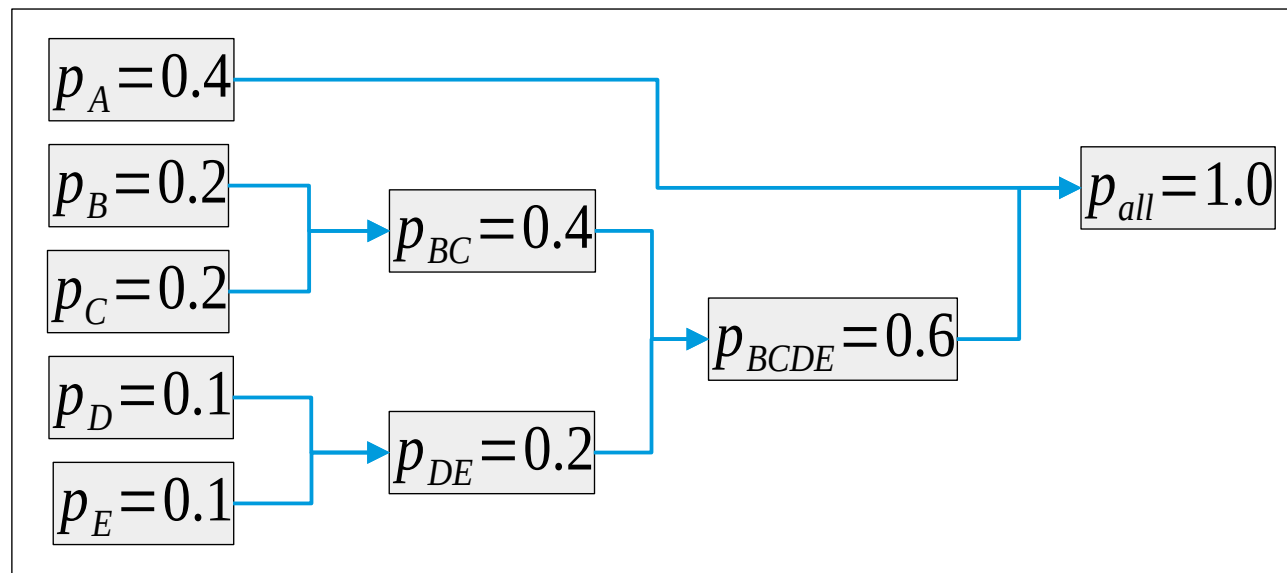
- average code length $H \leq \sum_{i=1}^n p_i l_i < H+1$

Shannon entropy and amount of information

- example for Huffman coding

- average questions:
 $0.4 + 3 \times 0.6 = 2.2$

- Shannon entropy:
2.12



Joint probability, entropy and mutual information

- if states depend on two variables, the joint probability $p(x, y)$
- for independent variables $p(x, y) = p(x)p(y)$
- the corresponding Shannon entropy

$$H(X, Y) = - \sum_{x \in X, y \in Y}^n p(x, y) \log_2 p(x, y) \rightarrow H(X) + H(Y)$$

- measure of independence: mutual information

$$I(X, Y) = H(X) + H(Y) - H(X, Y)$$



Joint probability, entropy and mutual information

- example: 2 bits

```
# Shannon entropy function
def SH(p):
    return -p*np.log2(p+1e-10)

pX = np.sum(p0, axis=1)
pY = np.sum(p0, axis=0)
pXY = np.reshape(p0, -1)
allH = []
for p in [pXY,pX,pY]:
    H = 0
    for pi in np.reshape(p, -1):
        H += SH(pi)
    allH.append(H)
IXY = allH[1] + allH[2] - allH[0]
```


Joint probability, entropy and mutual information

- example: 2 bits

```
# Shannon entropy function
```

```
def SH(p):
```

```
    return -p*np.log2(p+1e-10)
```

```
pX = np.sum(p0, axis=1)
```

```
pY = np.sum(p0, axis=0)
```

```
pXY = np.reshape(p0, -1)
```

```
allH = []
```

```
for p in [pXY,pX,pY]:
```

```
    H = 0
```

```
    for pi in np.reshape(p, -1):
```

```
        H += SH(pi)
```

```
    allH.append(H)
```

```
IXY = allH[1] + allH[2] - allH[0]
```

```
# example: independent case
```

```
p0=[[ 1/4, 1/4 ], [1/4, 1/4]]
```

```
pX = [1/2, 1/2]
```

```
pY = [1/2, 1/2]
```

```
# note that pXY = pX*pY
```

```
HX = 1
```

```
HY = 1
```

```
HXY = 2
```

```
IXY = 0
```

Joint probability, entropy and mutual information

- example: 2 bits

```
# Shannon entropy function
def SH(p):
    return -p*np.log2(p+1e-10)

p0 = np.array([[1/4, 1/4], [1/4, 1/4]])
pX = np.sum(p0, axis=1)
pY = np.sum(p0, axis=0)
pXY = np.reshape(p0, -1)
allH = []
for p in [pXY, pX, pY]:
    H = 0
    for pi in np.reshape(p, -1):
        H += SH(pi)
    allH.append(H)
IXY = allH[1] + allH[2] - allH[0]
```

example: independent case

$p0 = \begin{bmatrix} 1/4 & 1/4 \\ 1/4 & 1/4 \end{bmatrix}$

$pX = [1/2, 1/2]$

$pY = [1/2, 1/2]$

*# note that $p_{XY} = pX * pY$*

$HX = 1$

$HY = 1$

$HXY = 2$

$IXY = 0$

example: not independent case

$p0 = \begin{bmatrix} 1/3 & 1/6 \\ 1/4 & 1/4 \end{bmatrix}$

$pX = [1/2, 1/2]$

$pY = [7/12, 5/12]$

*# note that $p_{XY} \neq pX * pY$*

$HX = 1$

$HY = 0.98$

$HXY = 1.96$

$IXY = 0.02$

Probability of the observed sample

- we randomly choose case i with probability q_i
- what is the probability that we observe N_i events from class i ?
- logic of the result:

→ first we choose event 1, next event 2, etc. $\rightarrow \prod_{i=1}^n q_i^{N_i}$

→ number of such events is $\frac{N!}{N_1! N_2! \dots N_n!}$

→ probability $P = N! \prod_{i=1}^n \frac{q_i^{N_i}}{N_i!}$

Probability of the observed sample

- logarithm of the result in the $N \rightarrow \infty$ case

$$\log P = N \log N + \sum_{i=1}^n (N_i \log q_i - N_i \log N_i) = N \sum_{i=1}^n p_i \log \frac{q_i}{p_i}$$

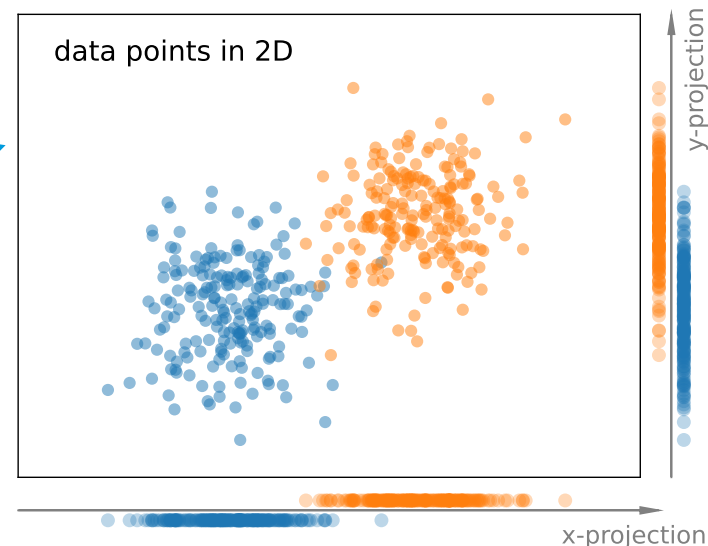
- Kullback-Leibler divergence: $D(p||q) = \frac{1}{N} \ln P$

$$D(p||q) = \sum_{i=1}^n p_i \ln \frac{q_i}{p_i}$$

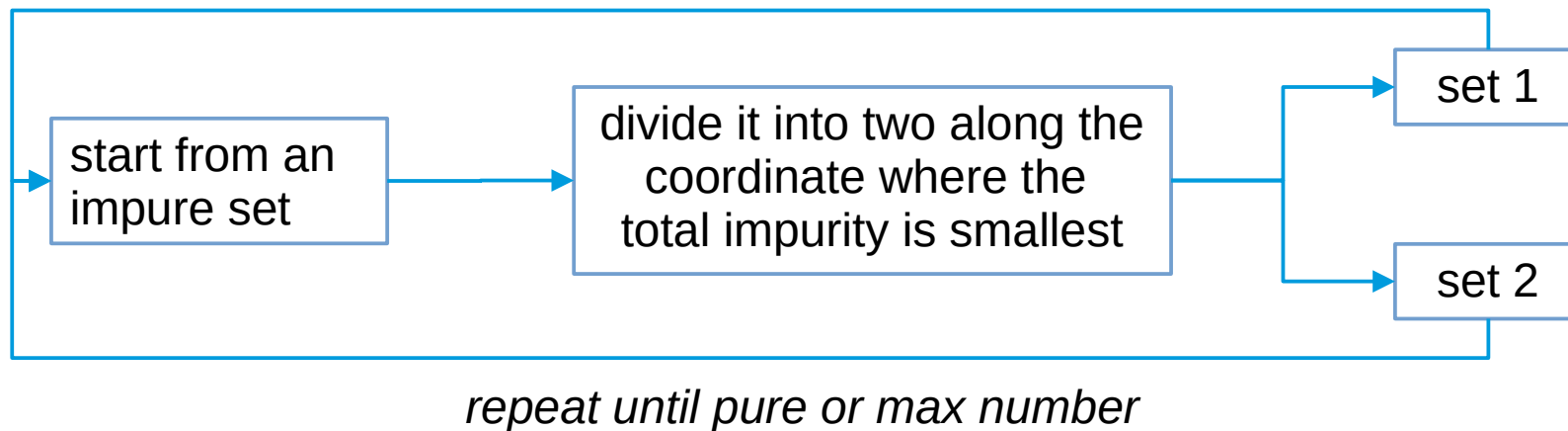
- interpretation: “distance” of probability distributions

Decision trees and ensemble improvements

- The **generalization to multiple dimensions** is not simple
 - in 1D the coordinates are unique
 - in multidimensional case we can optimize the coordination → features
 - simplest case: use original coordinates
 - usually the original coordinates are not equally useful
(c.f. *image* → *x coord separates more*)
 - choose that coordinate that leads to the smallest total impurity!



- Decision tree (DT)



- always converges, but may result in fragmented divisions

- Decision tree (DT) example

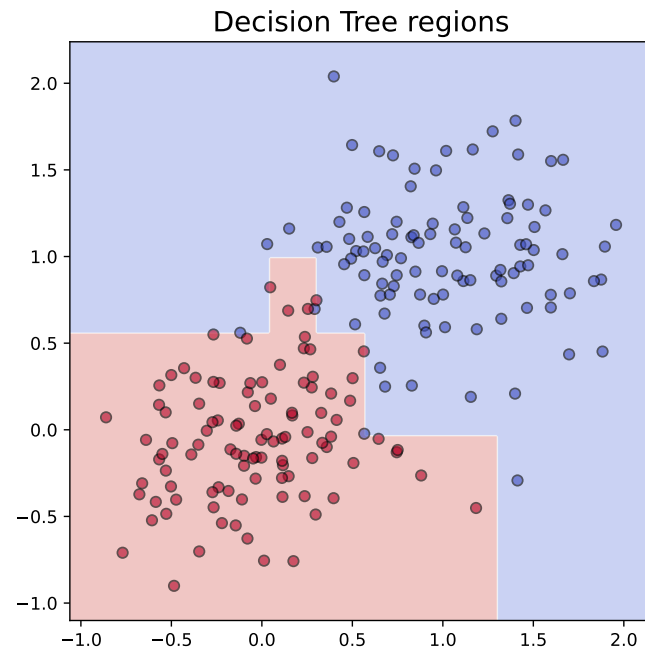
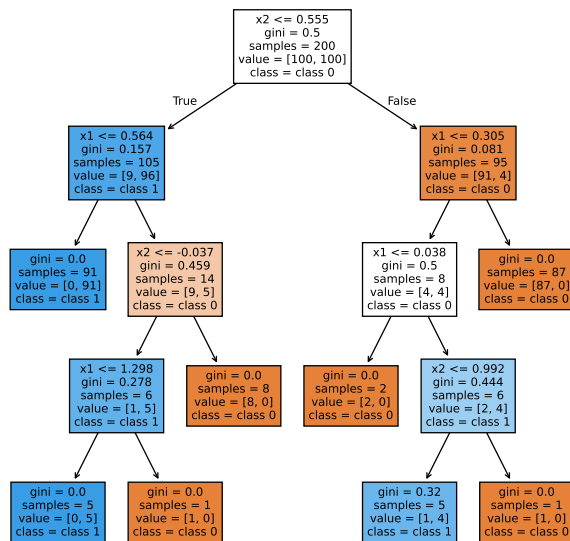
- two 2D sets

- pros:

- simple
 - interpretable

- cons:

- not robust enough
($\rightarrow$ *irregular shape*)



Tree based methods

- Decision tree (DT) example

 - two 2D sets

- pros:

 - simple

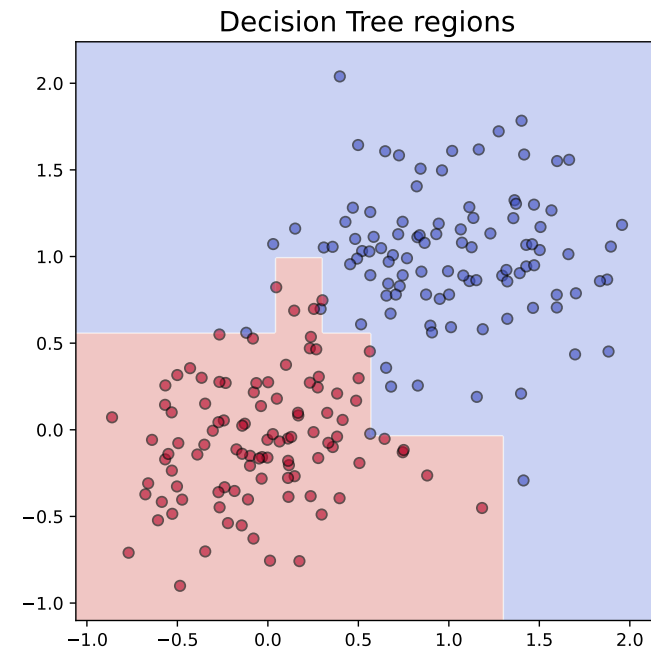
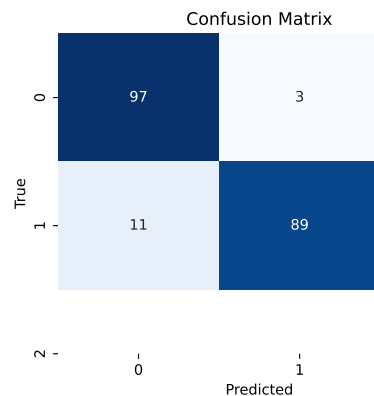
 - interpretable

- cons:

 - not robust enough

 - (→ *irregular shape*)
use fixed depth!

 - accuracy: *train: 100%, test: 92.5%*





- Decision tree (DT) technical solution in Python → [github](#)

```
# Decision Tree example  
from sklearn.tree import DecisionTreeClassifier  
from sklearn.metrics import confusion_matrix  
  
# 1. get or generate data and labels  
  
# 2. Fit a decision tree  
clf = DecisionTreeClassifier()  
clf.fit(data, labels)  
  
# 3. use for prediction  
pred_labels = clf.predict(other_data)  
  
#4. confusion matrix  
cm = confusion_matrix(labels, pred_labels)
```

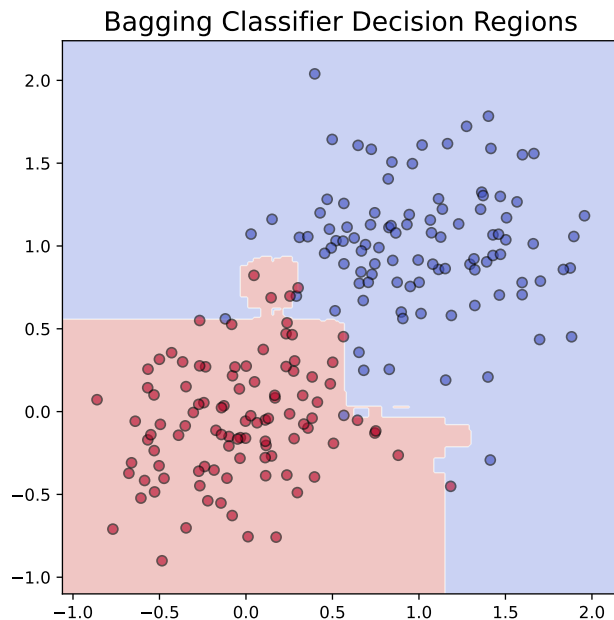


- Decision tree → too specific for a sample
- Possible solution: use ensemble of methods
- ways of improvement
 - bagging (Bootstrap AGGregatING) → bagging, random forest
 - boosting → adaptive boosting, gradient boosting
 - other methods → extra trees, isolated forests, oblique trees



- Idea:
 - train several models in slightly different datasets
 - each dataset contains the same number of data from the database, but some of them are missing, some appear multiple times
 - combine the results → regression: average, classification: majority vote
- Benefits:
 - lowers variance
 - increases robustness

- Code: [github](#)



Bagging example

```
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix
```

bagging based on decision tree

```
base_tree = DecisionTreeClassifier()
clf = BaggingClassifier(
    estimator=base_tree,
    n_estimators=50,           # number of trees
    max_samples=0.8,         # each tree sees 80% of data
    bootstrap=True,          # sampling with replacement
    random_state=42
)
clf.fit(X,y)
```

3. use for prediction

```
pred_labels = clf.predict(other_data)
```

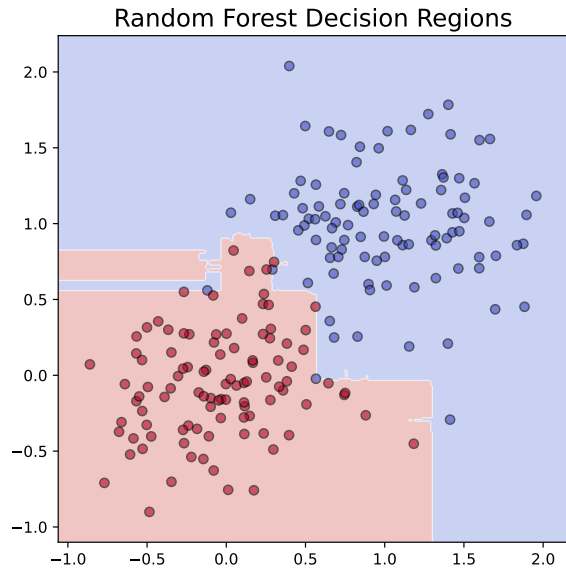
#4. confusion matrix

```
cm = confusion_matrix(labels, pred_labels)
```



- Idea: in addition vary the used parameters, too
 - train several models in slightly different datasets
 - each model uses selected data and selected parameters
 - combine the results → regression: average, classification: majority vote
- Benefits:
 - more robust than simple bagging
 - can be used to rank parameters by their significance, i.e. which contributed more to the final decision

- 2D case
- accuracy: 94%



```
# Random Forest example
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix
# bagging based on decision tree

# Train Random Forest
clf = RandomForestClassifier(
    n_estimators=100, # number of trees
    max_depth=None, # let trees grow fully
    random_state=42 # random generator seed
)
clf.fit(X, y)

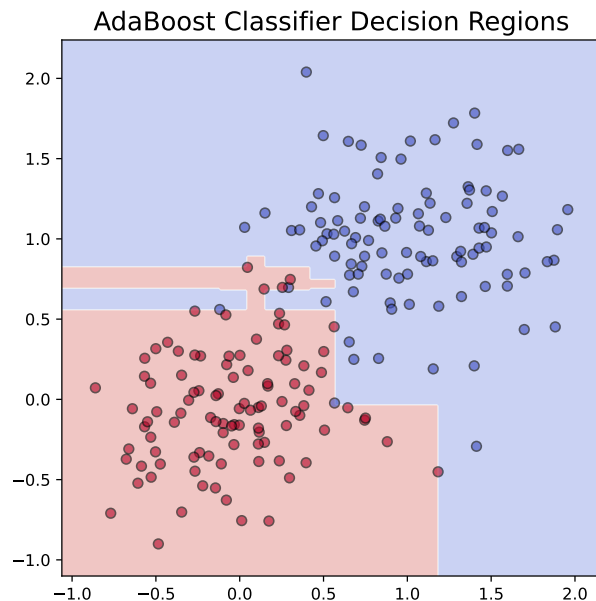
# 3. use for prediction
pred_labels = clf.predict(other_data)

#4. confusion matrix
cm = confusion_matrix(labels, pred_labels)
```



- Idea:
 - calculate a series of decision trees
 - each focuses on the points where the previous one fails
 - combine weighted sum of all of these models
- Adaptations: AdaBoost, Gradient Boosting (GBM), XGBoost, ...

- 2D case
- accuracy: 94%



AdaBoost example

```
from sklearn.metrics import confusion_matrix  
from sklearn.ensemble import AdaBoostClassifier
```

boosting based on decision tree

Train AdaBoost model

```
boost = AdaBoostClassifier(  
    estimator=DecisionTreeClassifier(max_depth=2),  
    n_estimators=50,  
    learning_rate=1.0,  
    random_state=42)  
boost.fit(X, y)
```

3. use for prediction

```
pred_labels = boost.predict(other_data)
```

#4. confusion matrix

```
cm = confusion_matrix(labels, pred_labels)
```

Estimate relevance of features



Relevance of features

- not all features contribute equally to the result – to reduce the number of features we shall assess their relevance
- idea: measure the change in the result when change the feature
- relevance measures (the most used ones)
 - ➔ feature importance (only for random forest)
 - ➔ permutation importance, feature masking
 - ➔ mutual information
 - ➔ in deep neural networks Gradient-weighted Class Activation Mapping (Grad-CAM)



- we will demonstrate the feature relevance measures in a 2-class 20 dimensional dataset with 2000 elements

```
# make multidimensional dataset  
from sklearn.datasets import make_classification  
from sklearn.model_selection import train_test_split  
  
ndim = 20  
Nsamples = 2000  
  
X, y = make_classification(n_samples=Nsamples, n_features=20,  
                           n_classes=2, random_state=42)  
  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,  
                                                    random_state=42)
```

- works only for random forest
- idea:
 - measure the impurity (e.g. Gini impurity) of all nodes before and after a decision
 - if we use a feature for a decision, measure the impurity decrease

$$\text{Importance}(f) = \sum_{\text{nodes using } f} \text{Impurity decrease at that node}$$

- average over the forest

$$\text{Importance}_{\text{forest}}(f) = \frac{1}{N_{\text{trees}}} \sum_{t=1}^{N_{\text{trees}}} \text{Importance}_t(f)$$

Feature importance

- example: create a 20 dimensional dataset with 2000 elements
- train a random forest classifier

```
# make mulitdimensional dataset  
from sklearn.ensemble import RandomForestClassifier  
  
# Creating a Random Forest classifier  
clf = RandomForestClassifier(n_estimators=100, random_state=42)  
  
# Training the classifier on the training data  
clf.fit(X_train, y_train)
```

- example: create a 20 dimensional dataset with 2000 elements
- train a random forest classifier
- evaluate the result

```
# evaluate result  
from sklearn.metrics import confusion_matrix  
  
# Making predictions on the testing data  
y_pred = clf.predict(X_test)  
  
# confusion matrix  
cm = confusion_matrix(y_test, y_pred)  
  
accuracy = np.trace(cm) / np.sum(cm)  
  
→ accuracy = 93.5%
```

- example: create a 20 dimensional dataset with 2000 elements
- train a random forest classifier
- evaluate the result
- get feature importance

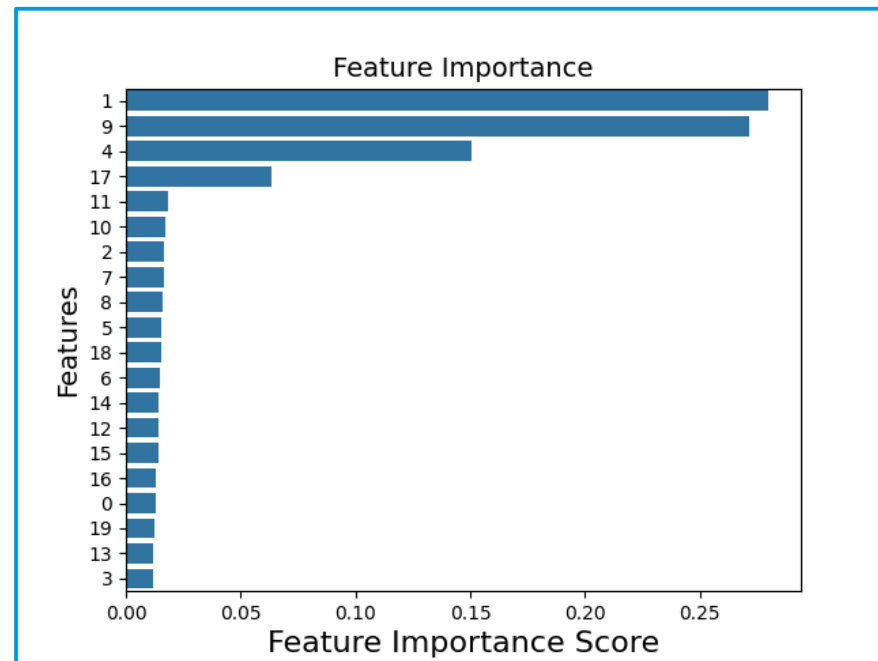
```
feature_importance = clf.feature_importances_  
  
→ np.array([0.01291599, 0.27999269, 0.01658218,  
0.01157727, 0.15054909, 0.01555601, 0.0150478 ,  
0.01637177, 0.01582075, 0.27170634, 0.01723006,  
0.01812296, 0.01401398, 0.01165928, 0.01404261,  
0.01398676, 0.01323008, 0.0636087 , 0.01551626,  
0.01246943])
```


Feature importance

- example: create a 20 dimensional dataset with 2000 elements
- train a random forest classifier
- evaluate the result
- get feature importance
- visualize it

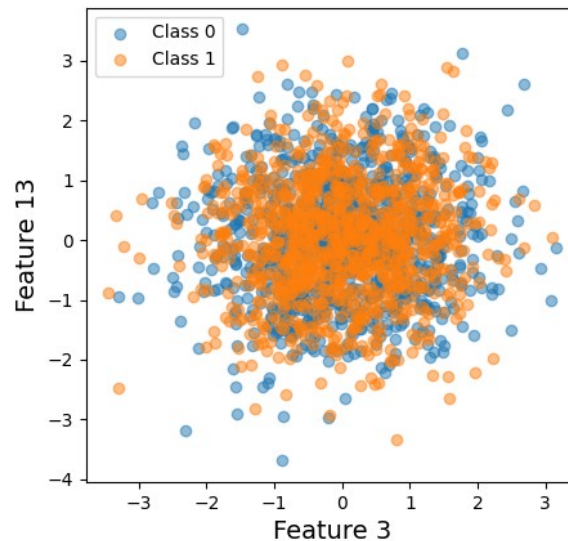
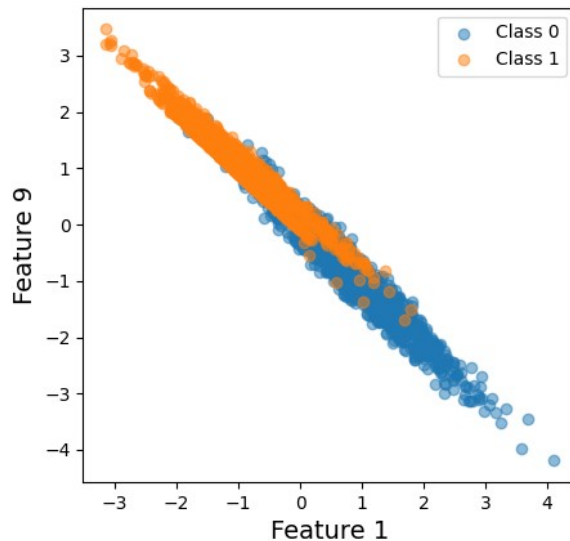
```
sorted_idx = np.argsort(feature_importance)[::-1]

plt.figure()
sns.barplot(
    x=feature_importance[sorted_idx],
    y=np.arange(len(sorted_idx)),
    orient='h')
plt.xticks(np.arange(len(sorted_idx)),sorted_idx)
```



Feature importance

- example: create a 20 dimensional dataset with 2000 elements
- most (1,9) and least (13,3) important features





- works for all methods
 - idea:
 - train the model, measure the performance – baseline
 - for each feature permute the values among samples
 - measure the performance again
 - relevance of the feature: drop in performance
- $$\text{Importance}(f) = \text{Score}_{\text{original}} - \text{Score}_{\text{permuted}}(f)$$
- instead permutation we can use a constant value (e.g. average), or a random value → mask that feature

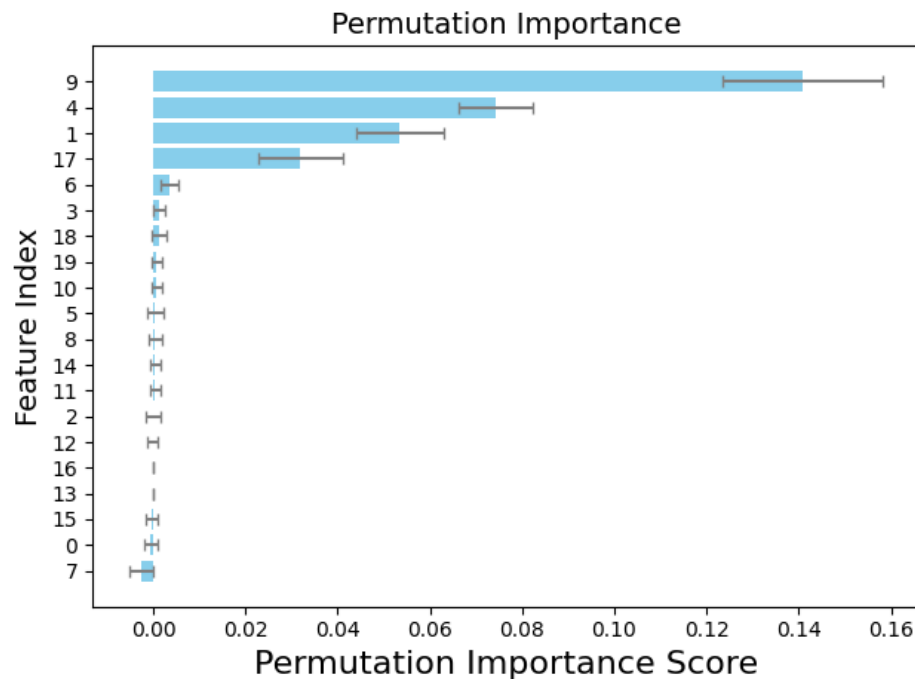


- get permutation importance in Python

```
# import package  
from sklearn.inspection import permutation_importance  
  
# use trained model clf to calculate importance  
result = permutation_importance(clf, X_test, y_test,  
                                n_repeats=10, random_state=42)  
  
# read mean and variance of the estimates  
importances = result.importances_mean  
std = result.importances_std
```

Permutation importance

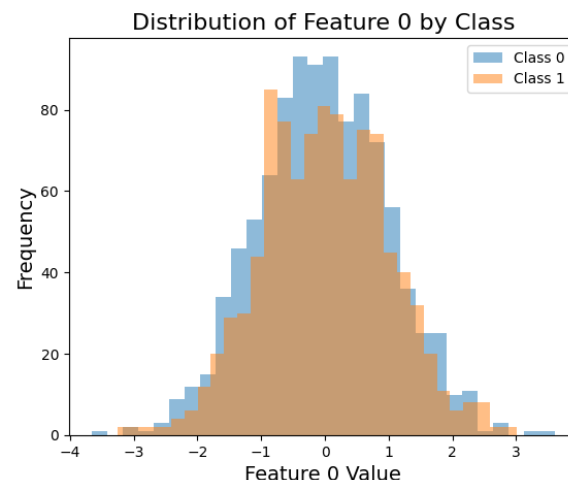
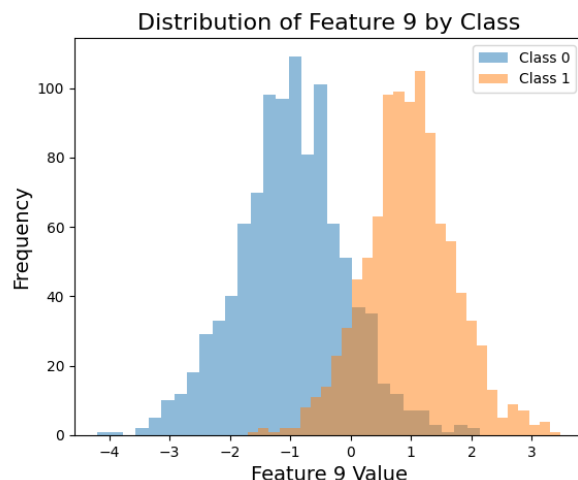
- get permutation importance in Python
- visualize the result
- same features are selected as before, but order is different



Permutation importance

- choose feature 9, and feature 0 to demonstrate the effect
- plot the distribution of the given feature, conditioned for the class (histogram)

$$P(\text{feature value} = f \mid \text{class})$$



- can be used for any models – for any two set, actually
- idea:
 - calculate the joint probability of two sets $p(x \in X, y \in Y)$
 - if x and y are independent, then the joint probability is the product

$$p(x, y) \xrightarrow{\text{independent}} p_X(x) p_Y(y)$$

- measure the dependence (mutual information) by

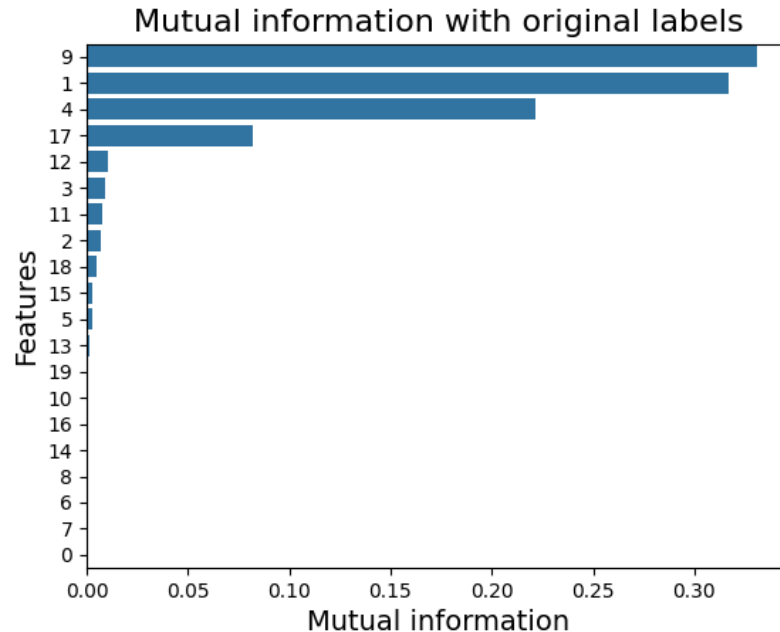
$$I(X, Y) = \sum_{x \in X, y \in Y} p(x, y) \ln \left(\frac{p(x, y)}{p_X(x) p_Y(y)} \right)$$

- can be used to measure mutual information between features and labels

- in Python there is a package to calculate mutual information
- the code

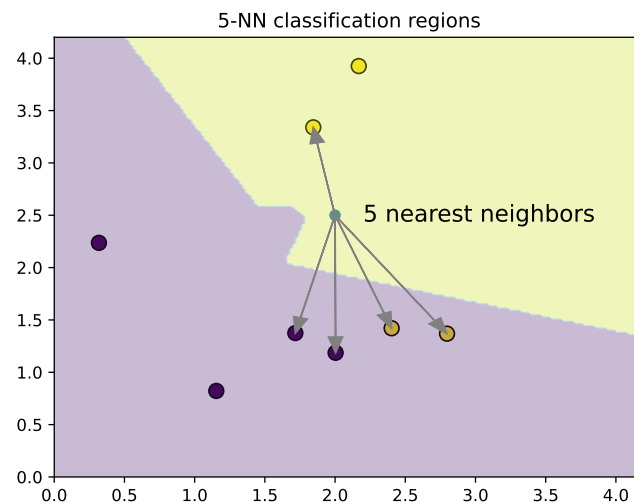
```
# import package  
from sklearn.feature_selection import mutual_info_classif, mutual_info_regression  
  
# we can use the original labels, or the model-predicted ones  
# characteristic for both the system and the trained model  
# y = clf.predict(X)  
  
mi_model = mutual_info_classif(X, y, random_state=42)  
mi_model /= np.sum(mi_model)
```


- in Python there is a package to calculate mutual information
- the code
- visualization
- similar result as before



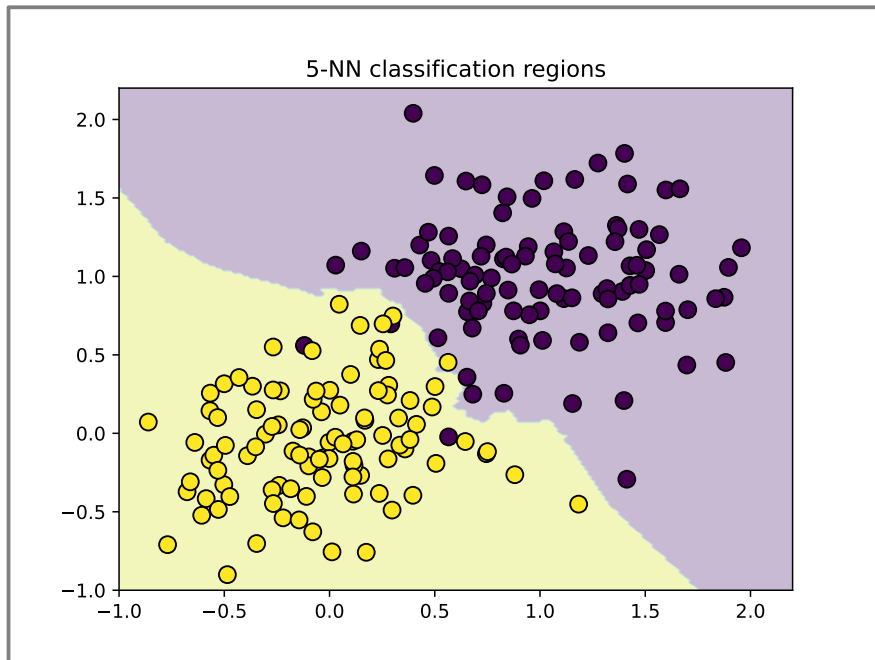
k-nearest neighbour (KNN) model

- a different idea: similar points are close in the feature space
- for a good guess on the class peek the neighbouring points
- algorithm:
 - keep all points in the training set
 - new point: compute all the distances
→ needs a distance measure
 - keep first k closest points
 - estimated class: majority vote



-  pros
 - simple, intuitive, no training phase
 - naturally handles multi-class problems
-  cons:
 - slow at prediction time (needs to compare with all samples)
 - sensitive to irrelevant features and scaling (c.f. distance measure)
 - struggles in high dimensions (“curse of dimensionality”)

- in code



```
# Random Forest example  
from sklearn import svm  
from sklearn.neighbors import KNeighborsClassifier
```

```
# Train a 3-NN classifier  
knn = KNeighborsClassifier(n_neighbors=5)  
knn.fit(X,y)
```

```
# Predict on the test set  
y_pred = knn.predict(X_test)
```

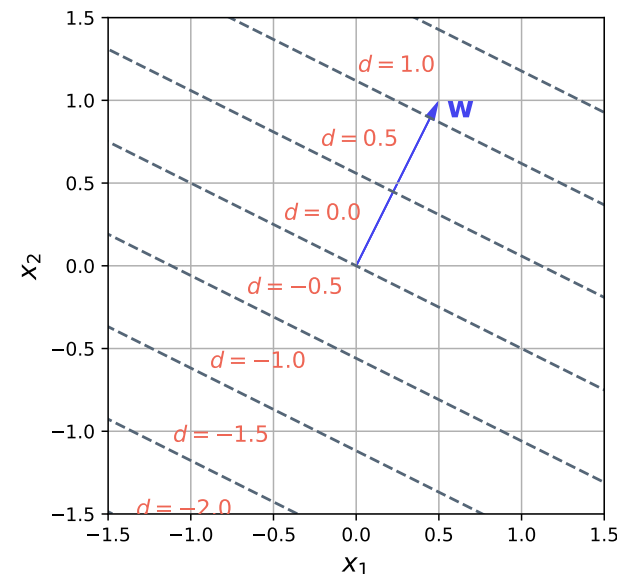
```
# Evaluate  
acc = accuracy_score(y_test, y_pred)
```

→ acc = 97.5%

Support Vector Machine

Support Vector Machine

- we can tailor features instead of the original ones
- simplest: linear combination
- logic:
 - ➔ define a w vector
 - ➔ the selective coordinate is $x \cdot w$
 - ➔ measures distance from hyperplanes orthogonal to w
 - ➔ try to separate the two sets using this feature
 - ➔ train to have the optimal w

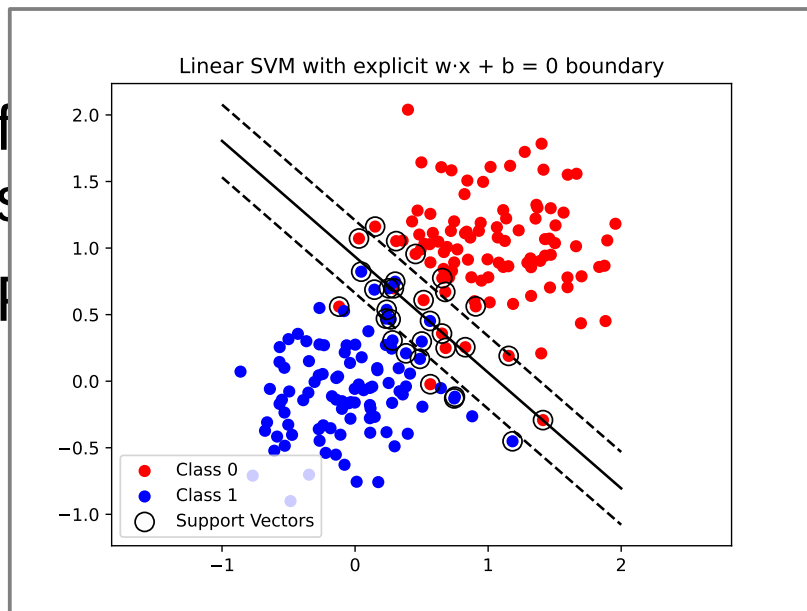




- straightforward optimization
 - map to 1D with the help of $x \rightarrow x \cdot w$
 - use 1D techniques (e.g. fitting probability function)
- for training one may use only the nearest points between the two sets (support vectors)
- Python tool: *svm*

- straightforward optimization

→ map to 1D with the help of $x \rightarrow$



Random Forest example

```
from sklearn import svm
```

```
from sklearn.metrics import accuracy_score
```

```
clf = svm.SVC(kernel='linear', C=1.0)
```

```
clf.fit(X, y)
```

🎮 Predict on the test set

```
y_pred = clf.predict(X_test)
```

📊 Evaluate

```
acc = accuracy_score(y_test, y_pred)
```

→ acc = 96.5%

- we can use also nonlinear combinations (kernel)
- most used: Gaussian kernel

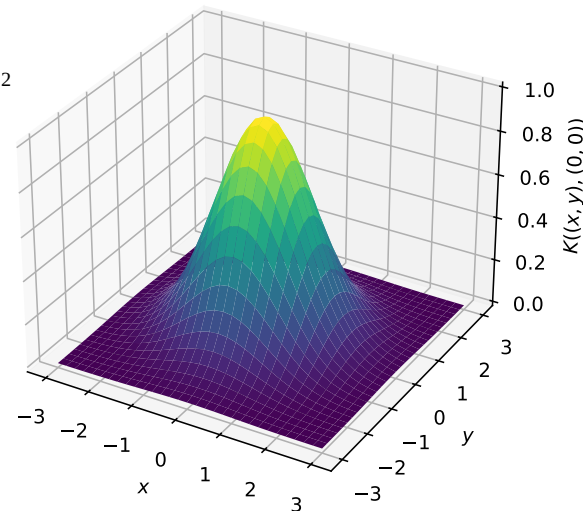
→ around each point we take a “bump” $x \rightarrow e^{-\gamma(x-x_0)^2}$

→ complete kernel is a linear combination
(using ± 1 as labels)

$$x \rightarrow \sum_n \alpha_n y_n e^{-\gamma(x-x_n)^2} + b$$

→ resulting surface is a landscape with hills and valleys

→ in practice it is enough to consider only a small number of points (support vectors) to represent the surface





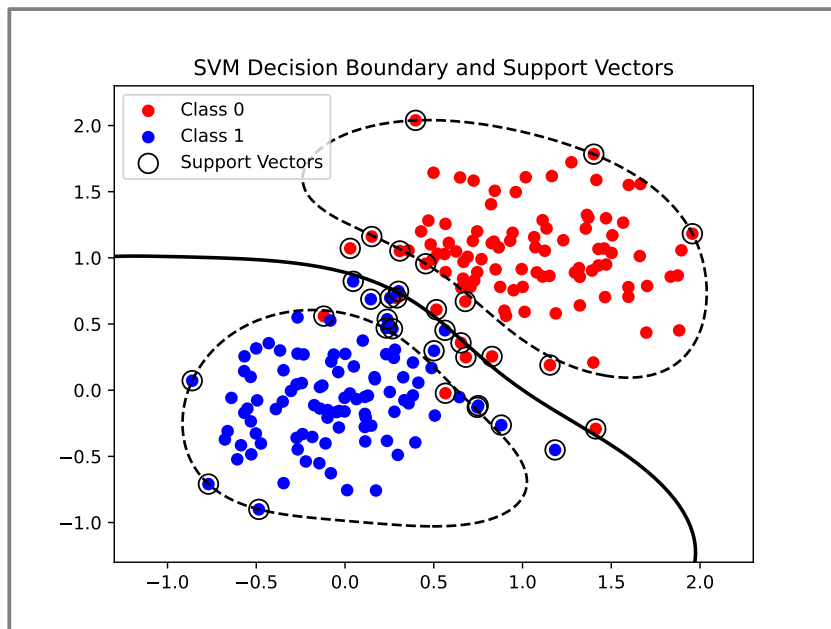
- we can use arbitrary kernel $x \rightarrow \sum_n \alpha_n y_n K(x_i, x; \Theta) + b$
- fitting and learning:
 - linear combination parameters α_n, b are learned using data
 - kernel parameters Θ are fixed in the learning process, and tuned for best performance (using grid search, cross validation)
 - Deep Neural Networks work in a similar manner, there the functional space is a nested (layered) function composition.

$$x \rightarrow W_n \sigma(W_{n-1} \sigma(\dots \sigma(W_1(x)) \dots))$$

- decision based on a single (although) complicated feature

Support Vector Machine

- in the code we should use the *'rbf'* kernel (*Radial Basis Function*)



Random Forest example

```
from sklearn import svm
from sklearn.metrics import accuracy_score
```

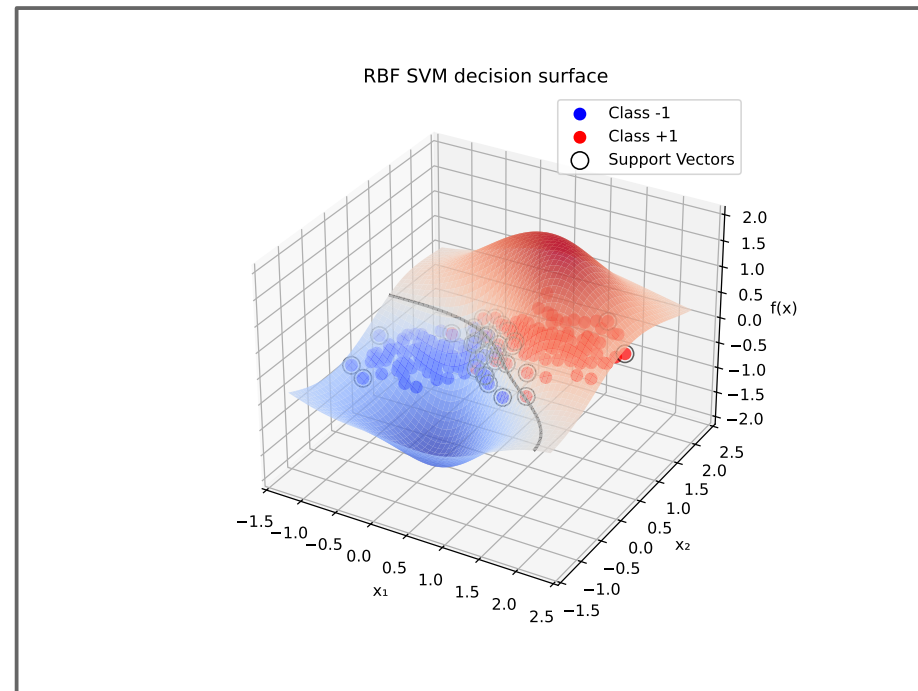
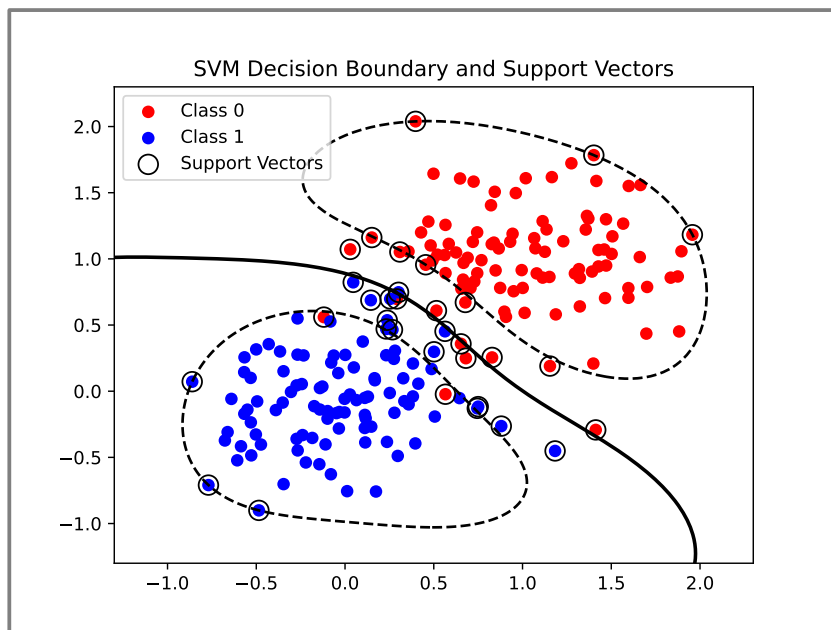
```
clf = svm.SVC(kernel='linear', C=1.0)
clf.fit(X, y)
```

```
# 🎩 Predict on the test set
y_pred = clf.predict(X_test)
```

```
# 📊 Evaluate
acc = accuracy_score(y_test, y_pred)
```

→ acc = 97.5%

- in the code we should use the 'rbf' kernel (*Radial Basis Function*)



Regression



Regression is optimization in a continuous parameter space

- Data model → feature optimization:
 - we need a parametric combination of the original variables
 - optimize with respect to some condition (c.f. SVM)
- Continuous value estimation:
 - the task can be to predict a continuous variable
 - classification can be based on probability distribution



task of regression

- we have data pairs $(X, Y) \rightarrow$ database
- we are looking for a model that $f(x; \omega) = y$, for all $x, y \in X \times Y$
 - we can not consider all possible functions (too large space)
 - fix a functional space (e.g. polynomials, Gaussians, etc.): $f \in F$
 - the variables ω select the actual function



task of regression

- there is no perfect model! → *philosophical problem ...*
 - not enough observation (data)
 - data do not contain all information
 - functional space does not cover all possible effects
- consequence: $f(x; \omega) = y$, for all $x, y \in X \times Y$ can not be satisfied
- what can be achieved: best representation, where the error is the smallest → what is “best”, and what is “error”?

task of regression – theoretical approach

- not enough information: $x \rightarrow y$ relation is not unique
- there is a probability to measure a *(data, label)* pair: $\rho(x, y) \in [0, 1]$
- for the complete dataset, assuming independent measurements

$$\rho(X, Y) = \prod_{n=1}^N \rho(x_n, y_n)$$

- for a given input there can be a lot of outputs: conditional probability (instead of a function)

$$\rho(Y | X) = \frac{\rho(X, Y)}{\rho(X)}$$

Bayes theorem

$$p(A | B) = \frac{p(A, B)}{p(B)}$$

task of regression – theoretical approach

- data model: $\rho(Y | X) \Rightarrow \rho(Y | X; \omega)$
- what is the best parameter choice? $\rightarrow$ *ensemble of worlds*
- define the “distribution of the parameters” using Bayes theorem

$$p(\omega | X, Y) = \frac{p(\omega, Y | X)}{p(Y | X)} = \frac{p(Y | \omega, X) p(\omega | X)}{p(Y | X)}$$

- as the function of the parameter $\rightarrow$ *likelihood function*
(assuming no restriction to possible parameter values)

$$L(\omega) = p(\omega | X, Y) \sim p(Y | \omega, X)$$

task of regression – theoretical approach

- best parameter value when $L(\omega) = p(\omega | X, Y)$ is maximal (*maximum likelihood*):
$$\frac{\partial L}{\partial \omega_i}(\omega_0) = 0$$
- *confidence levels*: $\Omega_p = \{\omega \mid L(\omega) \geq L_p = \text{constant}\} \rightarrow \text{hypersurfaces}$
- a confidence level can be characterized with the probability to be inside the volume
$$p = P(\omega \in \Omega_p)$$
 - for example: the probability that $\omega \in \Omega_{0.95}$ is 95% (according to $L(\omega)$)

task of regression – theoretical approach

$$N(\mu, C)(y) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- in practice we use Gaussian (*central limit theorem*)
 - assume that data points in the database are independent
 - data model gives the expected value
 - with unit correlation matrix → least square method (LSM)

$$\rho(Y | X; \omega) = \prod_{n=1}^N N(f(x_n; \omega), C_n)(y_n) \sim e^{-\frac{1}{2} \sum_{n=1}^N (y_n - f(x_n; \omega)) C_n^{-1} (y_n - f(x_n; \omega))} \Rightarrow e^{-\frac{1}{2} \chi^2}$$

- the likelihood $L(\omega) \sim e^{-\frac{1}{2} \chi^2(\omega)}$ → not Gaussian (except linear model)

task of regression – theoretical approach

$$N(\mu, C)(y) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)^T C^{-1}(y-\mu)}$$

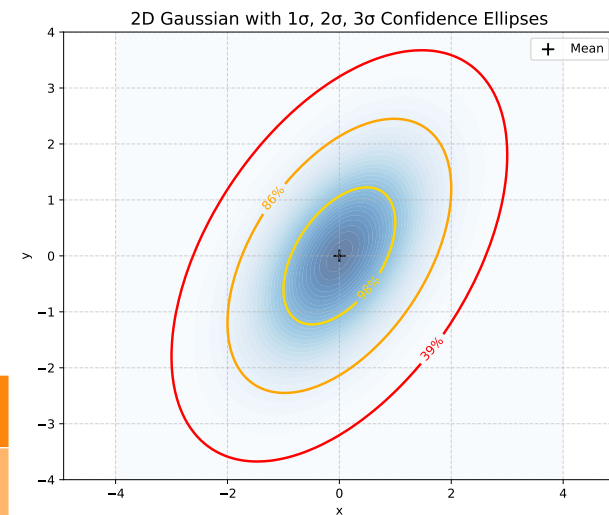
- expanding near the maximum likelihood → Gaussian with

$$\chi^2 = \chi_0^2 + (\omega - \omega_0)^T R^{-1} (\omega - \omega_0) + \dots$$

- Gaussian confidence levels

- the confidence surfaces are ellipsoids:
- confidence levels in one dimension $\chi^2 - \chi_0^2 = n$

sigma-level	1	2	3	4	5
confidence	68.27%	99.54%	99.73%	99.9937%	99.99994%



task of regression – theoretical approach

$$N(\mu, C)(y) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)^T C^{-1}(y-\mu)}$$

- Gaussian confidence levels and chi-squared distribution

$$p_d(\chi^2) = \frac{1}{2^{d/2} \Gamma(d/2)} (\chi^2)^{\frac{d}{2}-1} e^{-\frac{1}{2}\chi^2} d\chi^2$$

- probability to be inside $\chi^2 < r^2$ with the incomplete Gamma function

$$P(\chi^2 < r^2) = \int_0^{r^2} d\chi^2 p_d(\chi^2) = \frac{\gamma(d/2, r^2/2)}{\Gamma(d/2)}$$

$$\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt$$

$$\Gamma(a) = \gamma(a, \infty)$$

- in Python: `scipy.special.gammaincc`

$$\frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)^T C^{-1}(y-\mu)} d^d y = \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2}z^T z} d^d z = \frac{1}{2^{d/2} \Gamma(d/2)} (\chi^2)^{\frac{d}{2}-1} e^{-\frac{1}{2}\chi^2} d\chi^2$$

task of regression – theoretical approach

$$N(\mu, C)(y) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- calculation with Gaussian distribution: practical rules
 - assume y is Gaussian
 - introduce zero-mean variables $\xi = y - \mu$
 - expected values: $E(\xi) = 0$, $E(\xi_i \xi_j) = C_{ij} \Rightarrow E(\xi \xi) = C$
 - higher expected values are product of pairs!
e.g.:
$$E(\xi_i \xi_j \xi_k \xi_l) = C_{ij} C_{kl} + C_{ik} C_{jl} + C_{il} C_{jk}$$

task of regression – theoretical approach

$$N(\mu, C)(y) = \frac{1}{\sqrt{\det(2\pi C)}} e^{-\frac{1}{2}(y-\mu)C^{-1}(y-\mu)}$$

- hypothesis checking with chi-squared distribution:
 - assume Gaussian hypothesis and the data model are correct
 - treat the labels as independent random variables $y_n - f(x_n, \omega) \Rightarrow \xi_n$
 - compute expected value of chi-squared

$$E(\chi^2) = E\left(\sum_{n=1}^N \xi_n C_n^{-1} \xi_n\right) = \sum_{n=1}^N \sum_{i,j=1}^D (C_n^{-1})_{ij} E(\xi_{ni} \xi_{nj}) = N$$

- *chi-squared value over degrees of freedom is one*

Regression

Linear regression

simplest relation: linear function

- data are $(X, Y) = \{(x_n, y_n) \mid x_n \in \mathbb{R}^{N_x}, y_n \in \mathbb{R}^{N_y}, n = 1, 2, \dots, N\}$
- assume linear relation: $f(x; a, b) = ax + b$
- coefficient a is an matrix, b is a vector $[a] \Rightarrow N_x \times N_y, [b] \Rightarrow N_y$
- task to find the value (distribution or likelihood) of the coefficients
- simplest case, in lot of cases very powerful

one-dimensional case

- simplest: 1D linear regression, diagonal correlation matrix
- problem: data and labels are real numbers

$$(X, Y) = \{(x_n, y_n) \mid x_n, y_n \in \mathbb{R}, n = 1, 2, \dots, N\}$$

- assume there is a linear relation between the two: $f(x; a, b) = ax + b$

- chi-squared function $\chi^2 = \sum_{n=1}^N \frac{(ax_n + b - y_n)^2}{\sigma_n^2}$

- likelihood is Gaussian in a, b: $L(\omega) \sim e^{-\frac{1}{2}\chi^2(\omega)} = e^{-\sum_{n=1}^N \frac{(ax_n + b - y_n)^2}{2\sigma_n^2}}$



1D linear case

- computation: introduce new notations

- sum of inverse variances: $\sigma^2 = \frac{1}{N} \sum_{n=1}^N \frac{1}{\sigma_n^2}$

- averaging: $\langle u \rangle = \frac{\sum_n u_n / \sigma_n^2}{\sum_n 1 / \sigma_n^2} \Rightarrow \frac{1}{N} \sum_{n=1}^N u_n$

- correlation: $C_{uv} = \langle uv \rangle - \langle u \rangle \langle v \rangle = \langle (u - \langle u \rangle)(v - \langle v \rangle) \rangle$

- $\bar{b} = b + a \langle x \rangle - \langle y \rangle \Rightarrow y = \langle y \rangle + a(x - \langle x \rangle) + \bar{b}$

1D linear case

- computation: $\chi^2 = (a - C_{yx} C_{xx}^{-1}) \frac{N C_{xx}}{\sigma^2} (a - C_{yx} C_{xx}^{-1})^T + \frac{N \bar{b}^2}{N} + \text{constant}$

→ a and $\bar{b}$ are independent Gaussian variables

$$\mu_a = C_{yx} C_{xx}^{-1}, \quad C_a^{-1} = \frac{N C_{xx}}{\sigma^2}, \quad C_{\bar{b}}^{-1} = \frac{N}{\sigma^2}$$

→ the distribution of the variable $y = ax + b$ (with fixed x) is also Gaussian

$$\mu_y = \langle y \rangle + \mu_a (x - \langle x \rangle), \quad C_y = (x - \langle x \rangle) C_a (x - \langle x \rangle) + C_{\bar{b}}$$

1D linear case

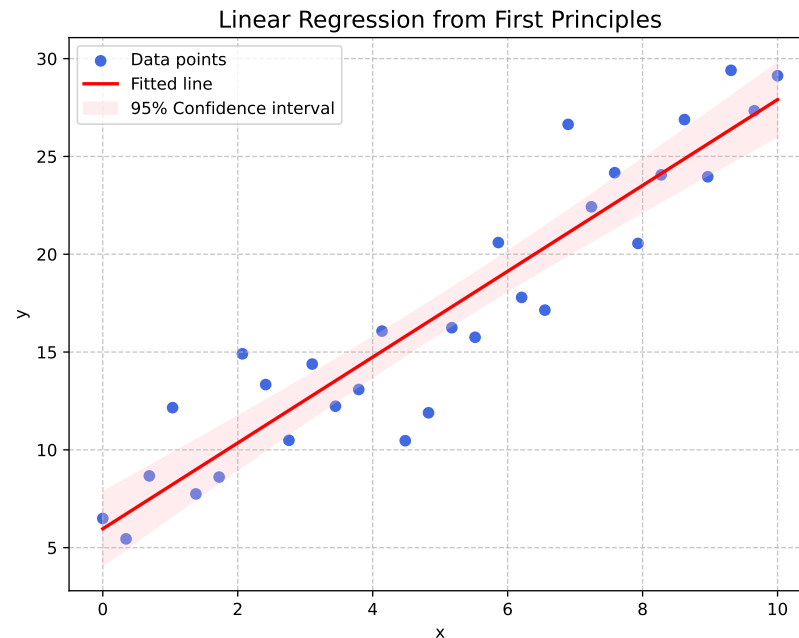
- interpretation:

- best slope value and variance

$$a = C_{yx} C_{xx}^{-1}, \quad \sigma_a^2 = \frac{1}{C_{xx} \sum_n 1/\sigma_n^2} \Rightarrow \frac{\sigma^2}{\sum_n (x_n - \bar{x})^2}$$

- chi-squared expected value

$$\frac{1}{N} \sum_n \frac{(y_n - ax_n - b)^2}{\sigma_n^2} = 0.81$$



higher dimensional linear case

- data are $(X, Y) = \{(x_n, y_n) \mid x_n \in \mathbb{R}^{N_x}, y_n \in \mathbb{R}^{N_y}, n = 1, 2, \dots, N\}$
- assume linear relation: $f(x; a, b) = ax + b$
- coefficient a is an matrix, b is a vector $[a] \Rightarrow N_x \times N_y, [b] \Rightarrow N_y$
- calculation and formulae remain the same

$$a = C_{yx} C_{xx}^{-1}, \quad C_a^{-1} = \frac{N}{\sigma^2} C_{xx}$$

higher dimensional linear case

- example: $N_x = N_y = 2, N_{data} = 100,$

$$a_{true} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad b_{true} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- then the result is

$$a_{data} = \begin{pmatrix} 0.99 & 2.03 \\ 3.06 & 3.96 \end{pmatrix}, \quad b_{data} = \begin{pmatrix} 0.98 \\ -0.98 \end{pmatrix}$$

```
# linear regression in 2D
```

```
Ndata = 100
```

```
a = np.array([[1, 2],[3, 4]])
```

```
b = np.array([1,-1])
```

```
x = np.random.uniform(0,1, size=(2, Ndata))
```

```
noise = np.random.normal(0,0.1, size=(2, Ndata))
```

```
y = a@x + noise + b[:,None]
```

```
barx = np.mean(x, axis=1)
```

```
bary = np.mean(y, axis=1)
```

```
Cxx = (x-barx[:,None])@(x-barx[:,None]).T/2
```

```
Cyx = (y-bary[:,None])@(x-barx[:,None]).T/2
```

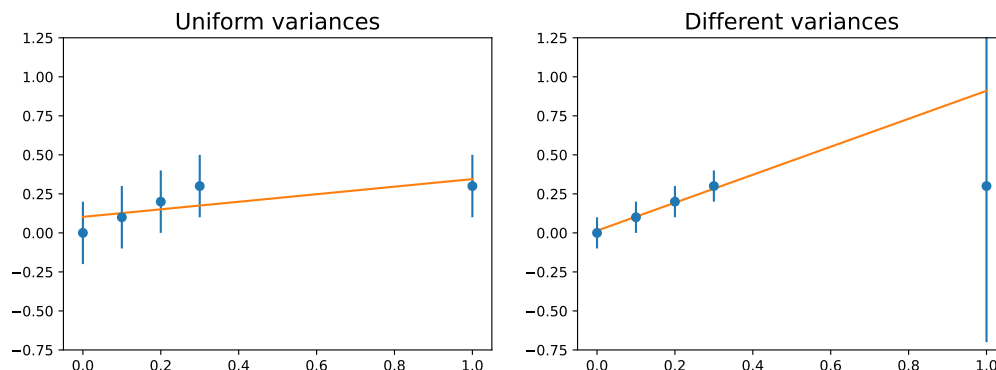
```
a1 = Cyx@np.linalg.inv(Cxx)
```

```
b1 = bary - a1@barx
```

```
omega1, b1
```

role of the individual variances

- effect of different variances → weight importance of points



- variance

- ➔ measurement (tedious)
- ➔ theoretical considerations (e.g. relative error is $\sigma \sim \sqrt{N}$)
- ➔ user expectations (e.g. some points are important)

role of the individual variances

- correlation matrix from measurements:

- assume that the data are i.i.d. normal $p(x) \sim \prod_{n=1}^N e^{-\frac{1}{2}(x_n - \mu)C^{-1}(x_n - \mu)}$

- distribution of $\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n$ is normal with $E(\bar{x}) = \mu \rightarrow$ estimate for mean

- mean of $\bar{C}_{ij} = \frac{1}{N-1} \sum_{n=1}^N (x_n - \bar{x})_i (x_n - \bar{x})_j \rightarrow E(\bar{C}_{ij}) = C_{ij} \rightarrow$ variance



fitting in a functional basis

- in lot of applications the task is to fit a function in a basis
- one-dimensional case: $f(t)$ function, $g_\alpha(t)$ basis, and we seek a_α :

$$f(t) = \sum_{\alpha=1}^{N_{basis}} a_\alpha g_\alpha(t)$$

- examples:
 - polynomial expansion: $g_\alpha(t) = t^\alpha \rightarrow$ linear fit uses 2D basis $g = (1, t)$
 - Fourier expansion: $g_\alpha(t) = e^{i\alpha t}$
 - Laplace transformation: $g_\alpha(t) = e^{-\alpha t}$

fitting in a functional basis

- the basis elements should be linearly independent, i.e.

$$0 = \sum_{\alpha=1}^{N_{basis}} a_{\alpha} g_{\alpha}(t) \Leftrightarrow a_{\alpha} = 0$$

- in practice we have data pairs $D = \{(t_n, y_n) \mid n=1, 2, \dots, N\}$

- solve $y_n \approx \sum_{\alpha=1}^{N_{basis}} a_{\alpha} g_{\alpha}(t_n)$, which means minimize $\chi^2 = \sum_{n=1}^{N_{data}} \frac{(y_n - \sum_{\alpha} a_{\alpha} g_{\alpha}(t_n))^2}{\sigma_n^2}$

- once we get the coefficients, we can interpolate: $f(t) \approx \sum_{\alpha=1}^{N_{basis}} a_{\alpha} g_{\alpha}(t)$
(prediction)

fitting in a functional basis

- computation (*more elegant than the linear case ...*)
 - introduce vector notation

$$X \in \mathbb{R}^{N \cdot N_{\text{basis}}}, \quad X_{n\alpha} = g_{\alpha}(t_n), \quad Y \in \mathbb{R}^N, \quad c \in \mathbb{R}^{N_{\text{basis}}} \Rightarrow \sum_{\alpha} a_{\alpha} g_{\alpha}(t_n) = X \cdot a$$

$$C \in \mathbb{R}^{N^2}, \quad C = \text{diagonal}(\sigma_1, \sigma_2, \dots, \sigma_N)$$

$$\chi^2 = \sum_{n=1}^{N_{\text{data}}} \frac{\left(Y_n - \sum_{\alpha} a_{\alpha} X_{n\alpha}\right)^2}{\sigma_n^2} = (X \cdot a - Y)^T C^{-1} (X \cdot a - Y)$$

fitting in a functional basis

- computation (*more elegant than the linear case ...*)
 - extremum: $\frac{\partial X^2}{\partial a} = 0 = X^T C^{-1} (X \cdot a - Y)$
 - solution: $a = (X^T C^{-1} X)^{-1} X^T C^{-1} Y$ ($\rightarrow$ c.f. linear case)
 - for uniform correlation matrix: $a = (X^T X)^{-1} X^T Y$

fitting in a functional basis

- example:

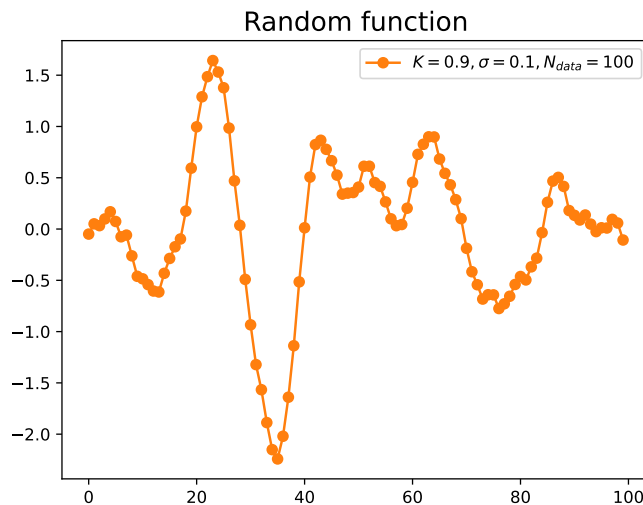
- create a random function, e.g. damped oscillator with noise

$$x_{n+1} = K(2x_n - x_{n-1} + \sigma \xi_n)$$

- choose functional basis, e.g. sin-cos

$$g_\alpha(t) = (\sin(\alpha t), \cos(\alpha t))$$

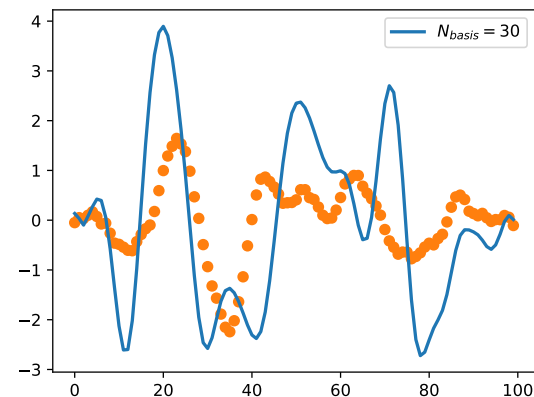
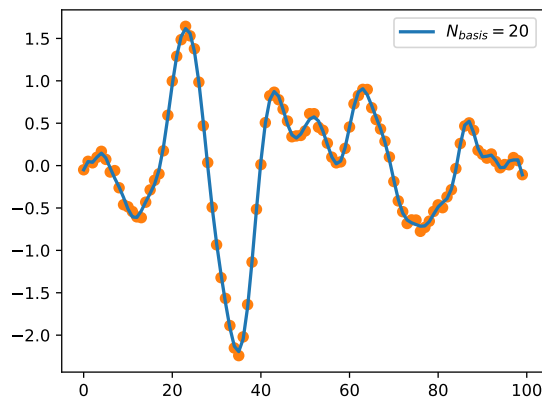
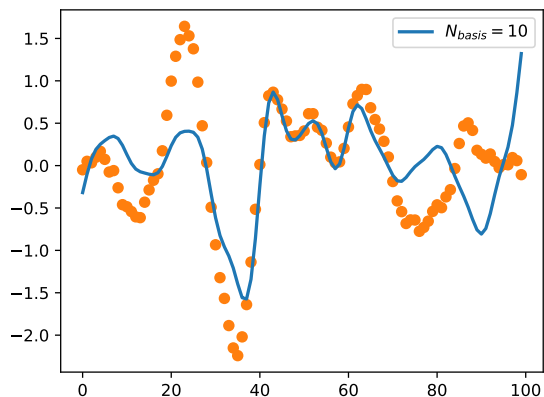
- use the fit formulae



fitting in a functional basis

- example:

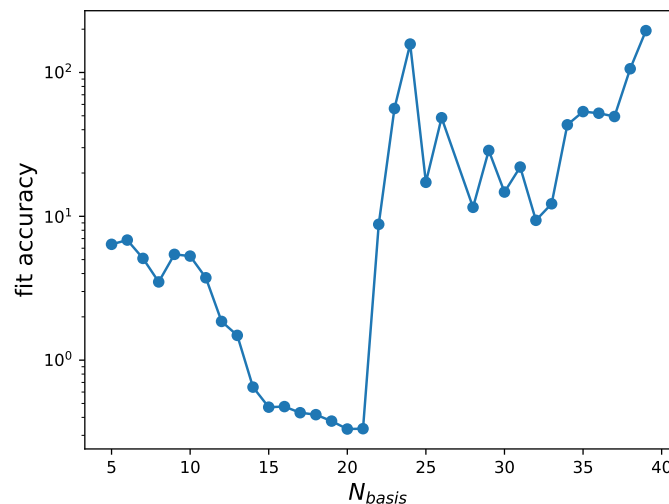
- ➡ result depends on the number of basis elements → non-monotonic!



fitting in a functional basis

- example:

- ➡ result depends on the number of basis elements → non-monotonic!





fitting in a functional basis

- what happened?
 - we have to invert a matrix: $a = (X^T X)^{-1} X^T Y$
 - what if there are small eigenvalues?

eigenvalues and eigenvectors in nutshell

- eigenvectors are stabile directions of a matrix

$$M v_n = \lambda_n v_n \quad V_{in} = (v_n)_i \Rightarrow MV = V \Lambda, \quad \Lambda = \text{diagonal}(\lambda_1, \lambda_2, \dots, \lambda_{N_{\text{basis}}})$$

- eigenvectors form a complete basis (*in most numerical cases*)

$$M = V \Lambda V^{-1} \Rightarrow M^{-1} = V \Lambda^{-1} V^{-1}$$

- matrix functions, defined with Laurent series

$$f(M) = \sum_k c_k M^k = \sum_k c_k (V \Lambda V^{-1})^k = V \left(\sum_k c_k \Lambda^k \right) V^{-1} = V f(\Lambda) V^{-1}$$



eigenvalues and eigenvectors in nutshell – special cases

- hermitian matrices: $M = M^+ = M^{*T} \rightarrow$ real symmetric matrices
 - eigenvalues are real
 - eigenvectors are orthogonal $V^+ V = V^+ V = 1, \quad M = V \Lambda V^+$
- $M = X^+ X$ matrices:
 - hermitian
 - eigenvalues are non-negative (positive definite)

$$\begin{aligned} M v_n = \lambda_n v_n &\Rightarrow v_n^+ M = \lambda_n^* v_n^+ \Rightarrow v_n^+ M v_m = \lambda_n^* v_n^+ M v_m = \lambda_m v_n^+ M v_m \\ X^+ X v_n = \lambda_n v_n &\Rightarrow \lambda_n = v_n^+ X^+ X v_n = |X v_n|^2 \end{aligned}$$

fitting in a functional basis

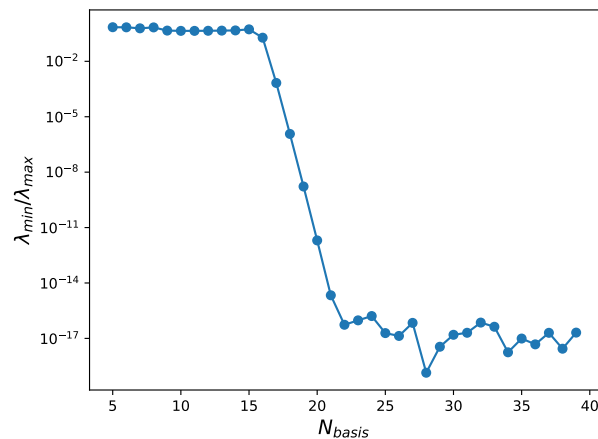
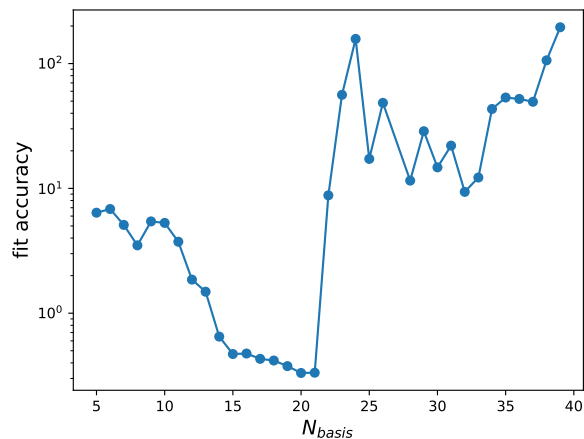
- what happened?
 - we have to invert a matrix: $a = (X^T X)^{-1} X^T Y$
 - what if there are small eigenvalues?
- a linear equation $Mx = y \Rightarrow x = M^{-1} y \rightarrow$ how reliable is it?
- add an eigenvector to the solution! $M(x + z v_0) = y + \lambda_0 z v_0$
- if numerical tolerance is $\epsilon \rightarrow |y + \lambda_0 z v_0| \approx |y| + \lambda_0 z < (1 + \epsilon)|y| \Rightarrow z < \frac{\epsilon}{\lambda} |y|$
- if $\lambda \ll \epsilon \Rightarrow z$ can be very large!

fitting in a functional basis

- typical value: $\epsilon \approx 10^{-12} - 10^{-17}$
 - measurement errors
 - numerical calculation errors
- ***ill-conditioned matrices:*** there are eigenvalues $\lambda \ll 10^{-15} \lambda_{max}$
the inverse is not reliable!
- very common in higher dimensions! → in ML or AI
- never use the naive linear equation solving formulae!

fitting in a functional basis

- in our example: for $N_{basis} \approx 21$ we reached Python's precision $\epsilon \sim 10^{-17}$
- after that value the fitting precision is drastically reduced





fitting in a functional basis

- what can we do with an ill-conditioned matrix?
 - pseudo-inverse
 - regularization

pseudo-inverse

- assume we can add the smallest eigenvector(s) with arbitrary coefficients
- choose that value that leads to the smallest length

$$x = \sum_n \alpha_n v_n \Rightarrow |x + c v_1|^2 = (\alpha_1 + c)^2 + \sum_{n=2}^{N_{basis}} \alpha_n^2 = \text{minimal} \Rightarrow c = -\alpha_1$$

- lesson: we should leave out all small eigenvectors!

- in formula:
$$M_{ij}^{-1} = \sum_{n=1}^{N_{basis}} \frac{1}{\lambda_n} v_{ni} v_{nj} \Rightarrow \sum_{\lambda_n > \epsilon \lambda_{max}}^{N_{basis}} \frac{1}{\lambda_n} v_{ni} v_{nj}$$

pseudo-inverse

- in Python code

```
# pseudoinverse fitting
```

```
nus = np.linspace(1,30,Nnu)
```

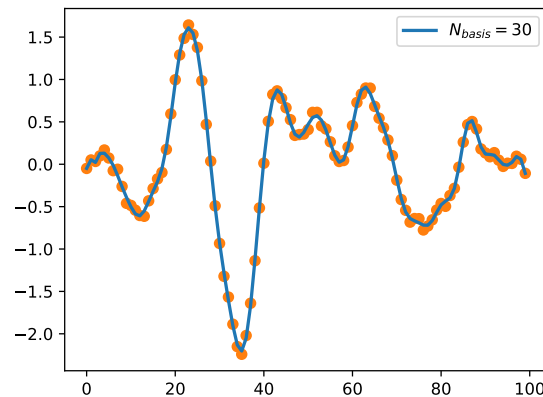
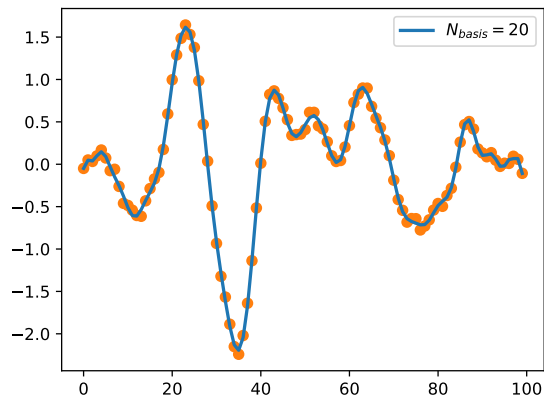
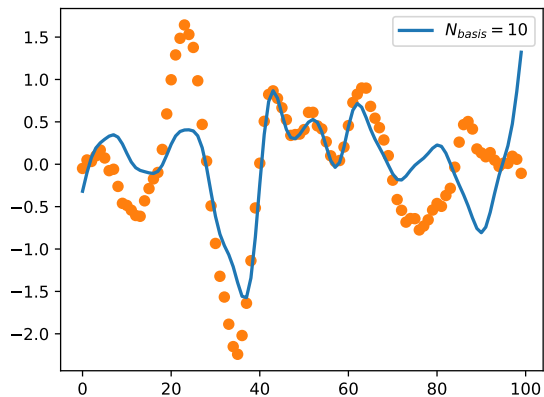
```
base1 = [[np.cos(np.pi*nu*n/Ndata) for n in range(Ndata)] for nu in nus]
```

```
base2 = [[np.sin(np.pi*nu*n/Ndata) for n in range(Ndata)] for nu in nus]
```

```
f = np.array(base1+base2).T
```

```
c = np.linalg.pinv(f.T@f)@f.T@y
```

```
plt.plot(f@c)
```





- pseudo-inverse works well, but closely connected to linear algebra
- more general method: **regularization**
- idea: add extra term to the loss function
 - stabilize result
 - spoils close connection to data
 - tradeoff: should use in a balanced manner
- most frequently used: L1, L2 regularization, MEM

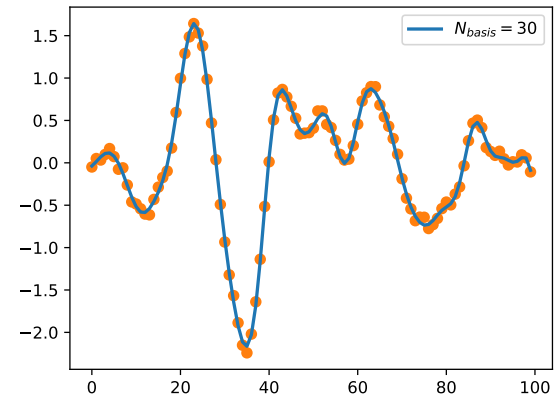
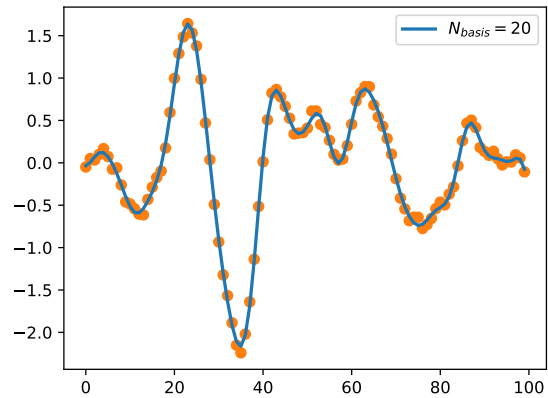
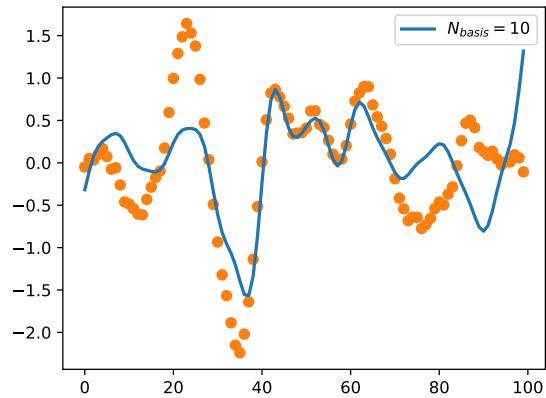
- L2 regularized loss

$$L = \sum_{n=1}^{N_{data}} \frac{(y_n - \sum_{\alpha} a_{\alpha} x_{n\alpha})^2}{\sigma_n^2} + \lambda \sum_{\alpha=1}^{N_{basis}} a_{\alpha}^2 \Rightarrow (X \cdot a - Y)^T C^{-1} (X \cdot a - Y) + \lambda a^T a$$

$$\frac{\partial L}{\partial a} = 0 = X^T C^{-1} (X \cdot a - Y) + \lambda a \Rightarrow a = (X^T C^{-1} X + \lambda)^{-1} X^T C^{-1} Y$$

- formally: add a small number to the matrix to be inverted
- works only for positive definite matrices (true in our case)

- L2 regularized loss results

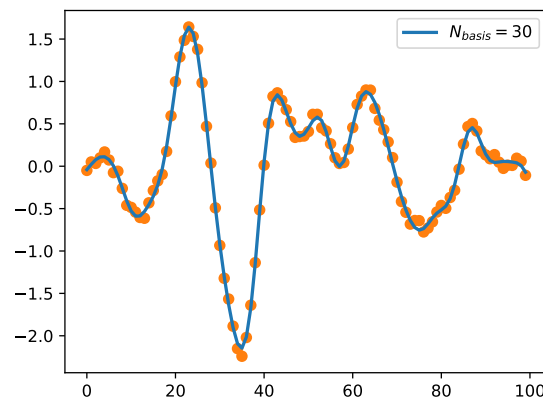
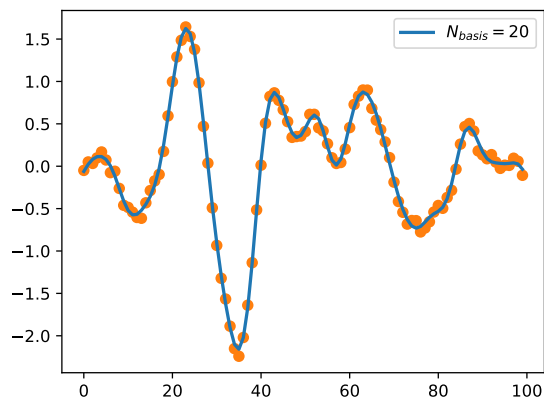
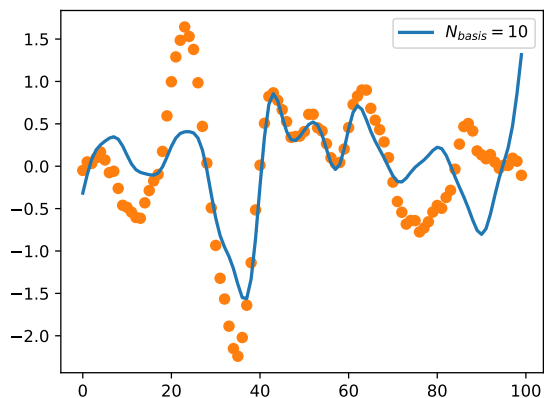


Linear regression

- L1 regularized loss – also for positive definite matrices

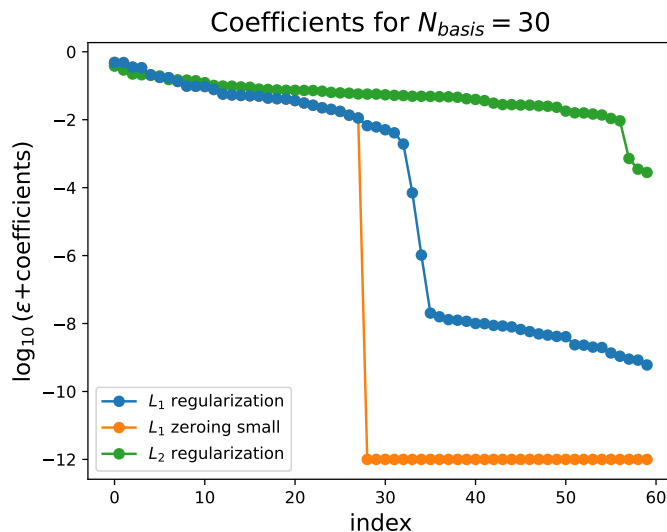
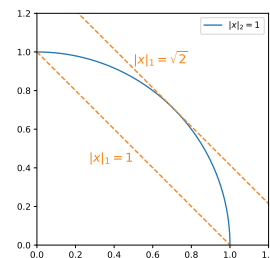
$$L = \sum_{n=1}^{N_{data}} \frac{\left(y_n - \sum_{\alpha} a_{\alpha} x_{n\alpha}\right)^2}{\sigma_n^2} + \lambda \sum_{\alpha=1}^{N_{basis}} |a_{\alpha}|$$

- minimization with numerical methods



Linear regression

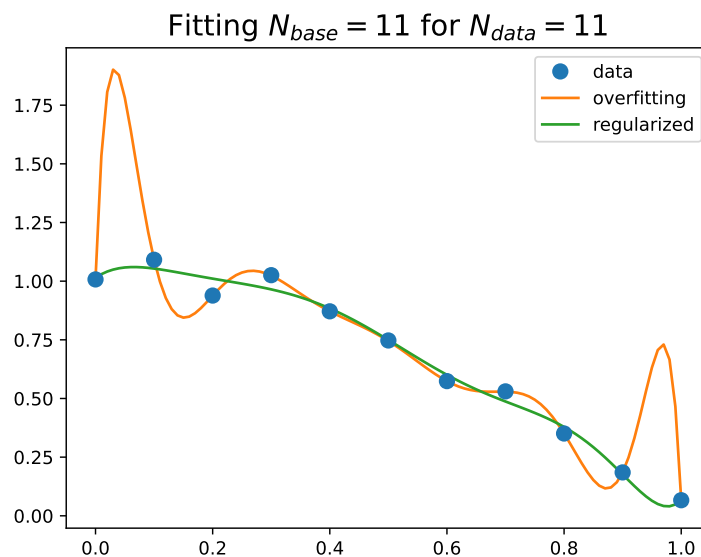
- L1 regularization benefit: likes small coefficients
- reason: $x^2 + y^2 = 1 \Rightarrow \min(|x| + |y|) \Rightarrow x=0, y=1$ or equivalent
- can be used to eliminate features $\rightarrow$ c.f. previous plot





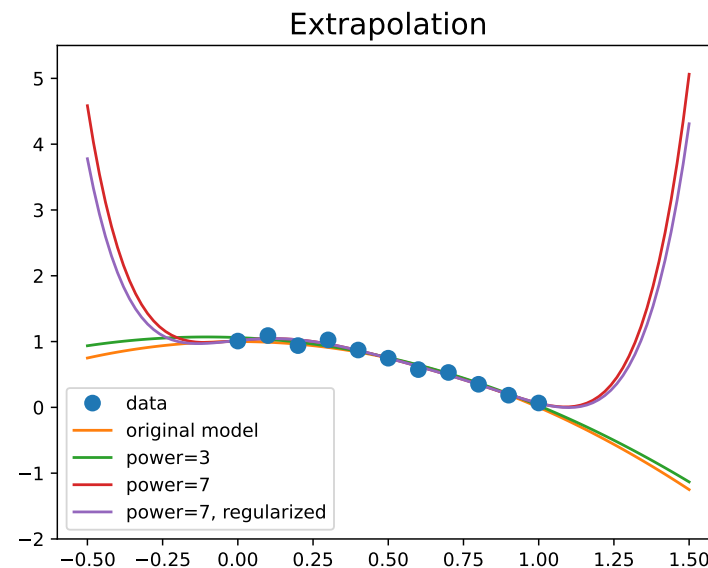
- interpolation vs. extrapolation
- after function fitting we can predict values
- data lay in $t_n \in T_{range} = [t_{min}, t_{max}]$
 - ➔ **interpolation**: prediction for $t \in T_{range}$
 - ➔ **extrapolation**: prediction for $t \notin T_{range}$
- dangers of interpolation → overfitting
- dangers of extrapolation → ...

- dangers of interpolation → **overfitting**
- example: fit polynomial basis to data points
- original model $y = 1 - x^2 + \xi$ with $N_{data} = 11$
- fitted model: $N_{basis} = 11$
with and or without regularization
- result:
 - ➔ exact fit, hits all points → bad interpolation
 - ➔ worse at the edges
 - ➔ regularization helps



Linear regression

- dangers of extrapolation → **avoid extrapolation**
- fitting lower and higher order polynomials
- results:
 - ➔ inside the data range all fits are correct
 - ➔ outside all fail
 - ➔ the more robust (fewer parameters) the better
 - ➔ regularization does not help!

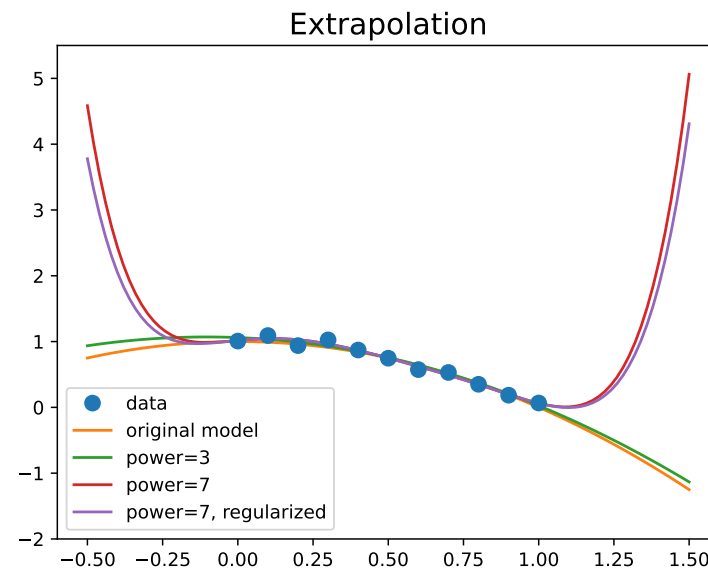


Linear regression

- dangers of extrapolation → **avoid extrapolation**
- fitting lower and higher order polynomials
- results:
 - ➔ inside the data range all fits are correct
 - ➔ outside all fail
 - ➔ the more robust (fewer parameters) the better
 - ➔ regularization does not help!



proceed and perish!





- controlling extrapolation
 - use bounded basis → avoid polynomials!
 - require asymptotic behaviour (i.e. function form at infinity)
→ transforms extrapolation to interpolation
 - use the model requiring the fewest parameters (robust modelling)
 - extend data in a sensible way

Linear representation of data distribution

Gaussian Mixture Models

- interconnected data are often sit in a compact region → ellipsoid
- find the ellipsoid containing most of the data! $p(x) \sim \prod_{n=1}^N e^{-\frac{1}{2}(x_n - \mu)C^{-1}(x_n - \mu)}$
$$L(\mu, C) = p(x | \mu, C) \text{ maximum} \Rightarrow \mu \approx \frac{1}{N} \sum_{n=1}^N x_n, \quad C_{ij} \approx \frac{1}{N-1} \sum_{n=1}^N (x_n - \bar{x})_i (x_n - \bar{x})_j$$
- probability level → “distance” $\chi^2 = (x_n - \mu)C^{-1}(x_n - \mu)$
- equidistance surfaces $\chi^2 = \text{constant} \rightarrow$ ellipsoids
- generate points: $x = \mu + r \sum_{a=1}^{\dim} \alpha_a \sqrt{\lambda_a} v_a \rightarrow \sum_{a=1}^{\dim} \alpha_a^2 = 1, \quad C v_a = \lambda_a v_a$

Gaussian Mixture Models

- technical slide: equipotential surfaces from correlation matrix
- we want to satisfy $\chi^2 = (x - \mu) C^{-1} (x - \mu) = r^2$
- eigensystem $C v_a = \lambda_a v_a$, eigenvectors orthogonal
- ansatz: $x = \mu + r \sum_{a=1}^{\dim} \alpha_a \sqrt{\lambda_a} v_a \Rightarrow x - \mu = r \sum_{a=1}^{\dim} \alpha_a \sqrt{\lambda_a} v_a$
- verify: $\chi^2 = r^2 \sum_{a,b=1}^{\dim} (\alpha_a \sqrt{\lambda_a}) (\alpha_b \sqrt{\lambda_b}) v_a C^{-1} v_b = r^2 \sum_{a,b=1}^{\dim} (\alpha_a \sqrt{\lambda_a}) (\alpha_b \sqrt{\lambda_b}) \lambda_a \delta_{ab} = r^2$

- technical slide: generate data with given μ, C

- data with normal distribution: ξ

- transform it $\eta = \sqrt{C} \xi + \mu$

- check:

$$E(\eta) = \mu$$

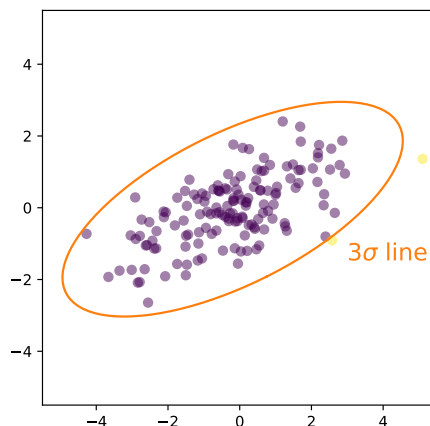
$$E((\eta - \mu) \otimes (\eta - \mu)) = \sqrt{C} E(\xi \otimes \xi) \sqrt{C} = C$$

- for 2D data we can directly specify the direction: $ev = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}$

```
# generating Gaussian with given mu and C  
def generate_Gaussian(mu, C, Ndata):  
    dim=mu.shape[0]  
    xi = np.random.normal(0,1, (dim, Ndata))  
    lam,ev =np.linalg.eigh(C)  
    sqrt_C = ev@np.diag(np.sqrt(lam))@ev.T  
    return sqrt_C@xi + mu
```

Gaussian Mixture Models

- example: 2D points $C = \begin{pmatrix} 2.5 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow \lambda = (0.5, 3), \text{ ev} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$
 - generate points
 - numerically determine correlation matrix and eigensystem
 - draw ellipses containing 99% of data → about 3σ in 2D



Gaussian Mixture Models

- if we have several classes → Naive Bayes models
 - assume that each class are independent and points lie in an ellipsoid → can be represented by Gaussian distribution
 - determine the probability to belong to each classes separately

```
# Naive Bayes classification
```

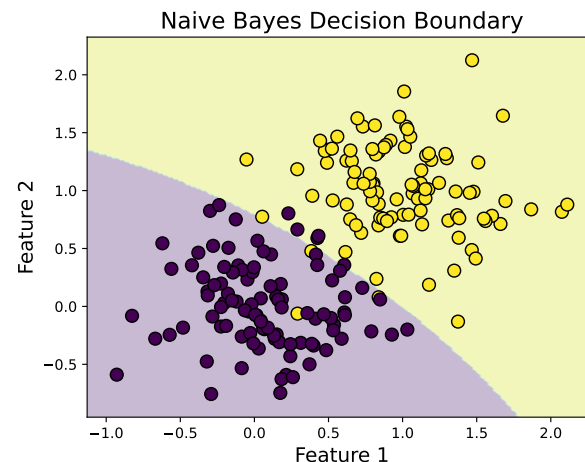
```
from sklearn.naive_bayes import GaussianNB
```

```
nb = GaussianNB()
```

```
nb.fit(X_train, y_train)
```

```
y_pred = nb.predict(X_test)
```

```
→ accuracy: 96%
```



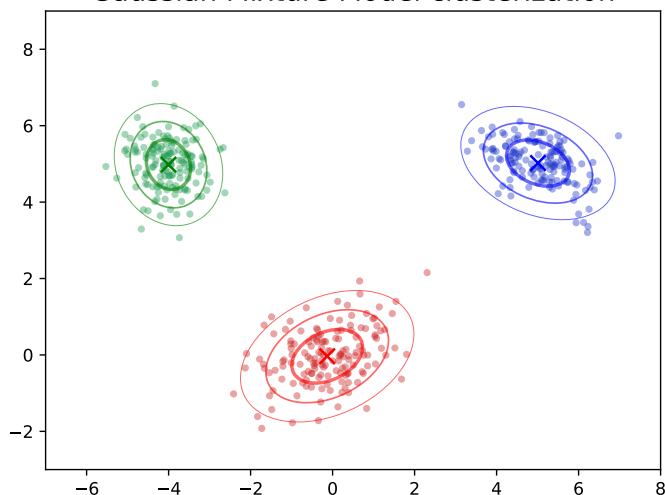
- unsupervised generalization → GMM
 - assume the probability distribution is a **sum** of Gaussian terms
 - fix the number of clusters → *do not provide labels (unsupervised)*
 - find the best parameters by maximizing the likelihood

$$L(\pi, \mu, C | x) = p(x | \pi, \mu, C) = \sum_{k=1}^K \pi_k N(x; \mu_k, C_k)$$

- we need convex combinations $\pi_k \in [0, 1], \sum_{k=1}^K \pi_k = 1$
- use Expectation–Maximization (EM) algorithm

- example: clustering in 2D

Gaussian Mixture Model clusterization

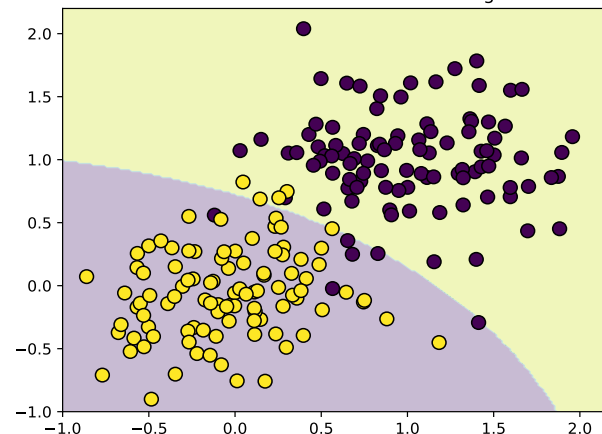


```
# Gaussian Mixture Models
```

```
from sklearn.mixture import GaussianMixture
```

```
gmm = GaussianMixture(n_components=2, covariance_type='full',  
random_state=0, n_init=10)gmm.fit(X)
```

Gaussian Mixture Models decision regions



- technical slide: the Expectation–Maximization (EM) algorithm

- start from an initial guess π, μ, C

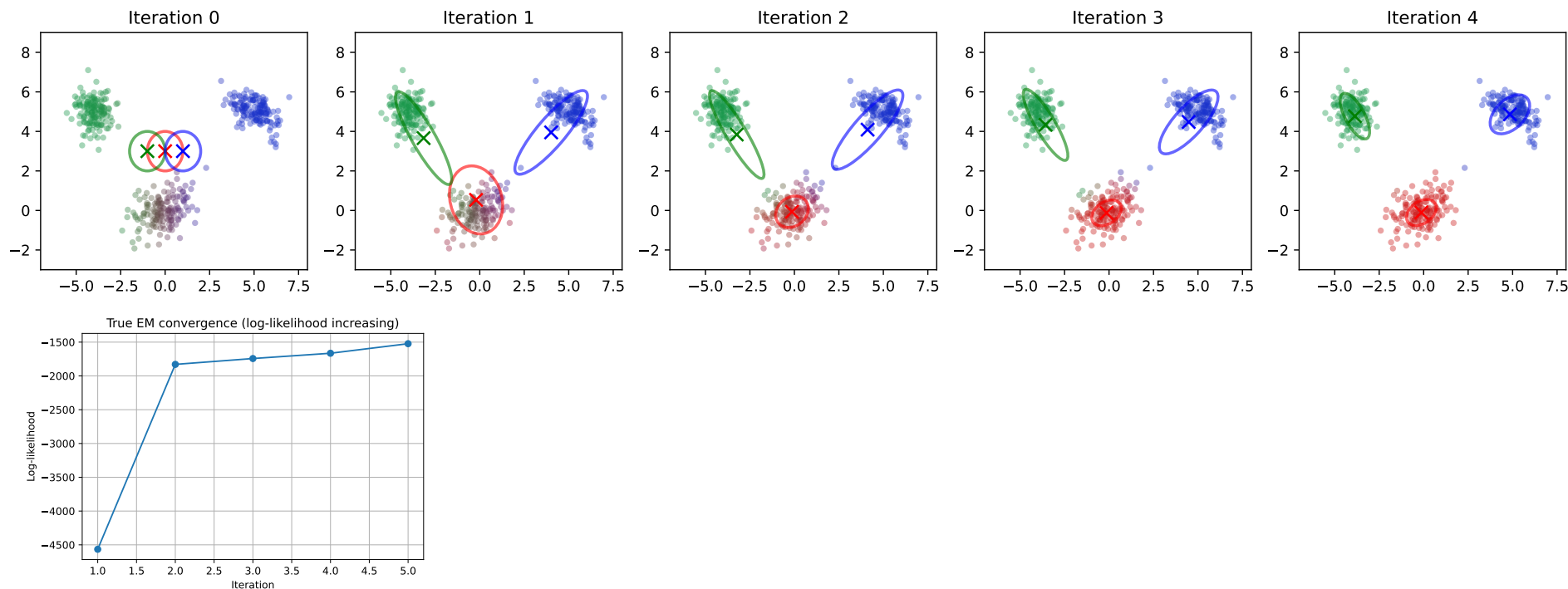
- compute “responsibilities”
(closest component weight) $r_{ik} = \frac{\pi_k N(x_i; \mu_k, C_k)}{\sum_j \pi_j N(x_i; \mu_j, C_j)}$

- redefine parameters
(attract to closest cluster)

$$N_k = \sum_i r_{ik}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{1}{N_k} \sum_i x_i r_{ik}, \quad C_k = \frac{1}{N_k} \sum_i (x_i - \mu_k)(x_i - \mu_k)^T$$

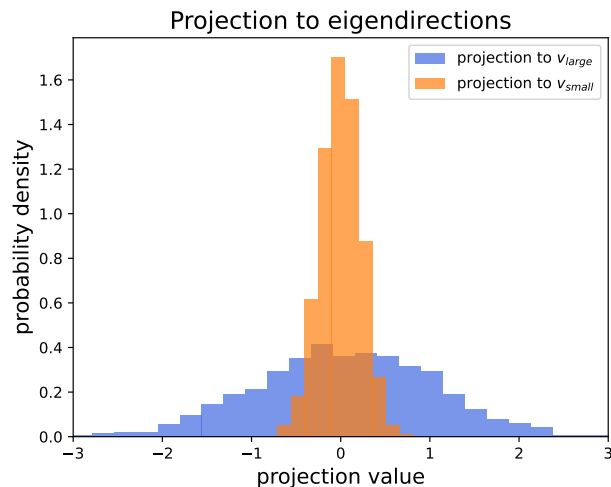
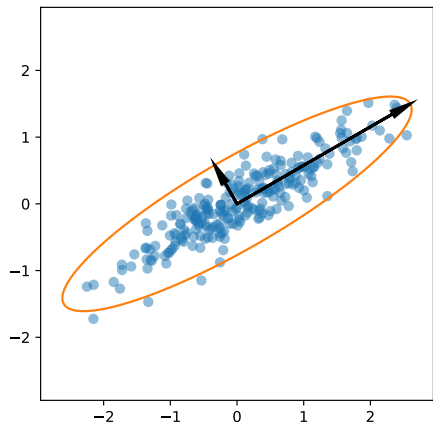
- repeat until converges

- technical slide: the Expectation–Maximization (EM) algorithm

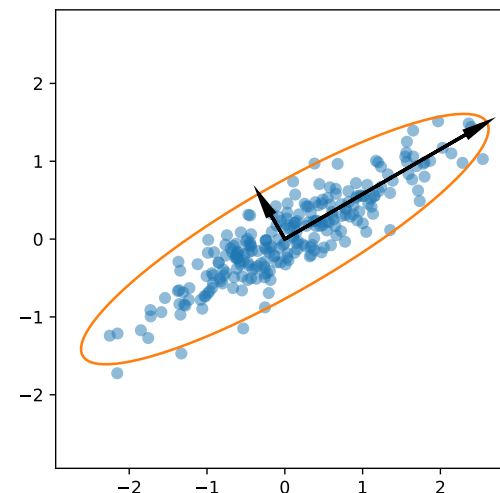


Principal Component Analysis (PCA)

- choose an orthonormal coordinate system aligned with the data!
- we fitted an ellipsoid $\rightarrow$ find the eigenvectors
 - $\rightarrow$ smallest eigenvalues: data vary the least
 - $\rightarrow$ largest eigenvalues: data vary the most



- PCA: keep only largest eigenvalues
 - ➔ compression → descriptive features
 - ➔ noise reduction
 - ➔ representation of the largest changes
- small eigenvalues:
 - ➔ approximately constant for all data in a class
 - ➔ classification → selective features
 - ➔ laws



- technical slide: compression with PCA

$$\text{data} = \{x_n \in \mathbb{R}^d \mid n=1, 2, \dots, N\} \quad X_{ni} = (x_n)_i$$

$$\mu_i = \frac{1}{N} \sum_n X_{ni}, \quad \bar{X}_{ni} = X_{ni} - \mu_i \quad C = \bar{X}^T \bar{X}$$

$$C v_a = \lambda_a v_a, \quad |v_a| = 1 \Rightarrow E((\bar{X} \cdot v_a)^2) = E(v_a^T \bar{X}^T \bar{X} v_a) = E(v_a C v_a) = \lambda_a$$

- eigenvalue = projection value squared
- PCA values: $Y_{na} = v_a^T \bar{X}_n$ for the largest eigenvalues $\rightarrow$ fewer data

Principal Component Analysis (PCA)

- technical slide: data reconstruction with PCA
- most information is in the subspace of largest eigenvalues

$$\bar{x}_n = \sum_{a=1}^d v_a (v_a^T \cdot \bar{x}_n) \approx \sum_{\lambda_a / \lambda_{\max} > r_{\min}} v_a (v_a^T \bar{x}_n) = \sum_{\lambda_a / \lambda_{\max} > r_{\min}} Y_{na} v_a$$

- therefore $x_n = \sum_{\lambda_a / \lambda_{\max} > r_{\min}} Y_{na} v_a + \mu_i$

